

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

GRADUATION THESIS

Automatic Product Classification According to the NICE Classification System

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Program: Data Science

Supervisor: Doctor Nguyễn Kiêm Hiếu

Department: Computer Science

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ABSTRACT

Trademark classification across multiple languages under the NICE Classification system plays a crucial role in safeguarding intellectual property in a globalized environment, yet the task remains highly challenging due to the large number of classes, the prevalence of domain-specific and ambiguous product descriptions, and the reliance on manual examination, which is both time-consuming and prone to errors. Automating this process using machine learning poses significant challenges, particularly due to severe class imbalance across the 45 NICE classes and the linguistic diversity of trademark descriptions. While pretrained multilingual models like Bidirectional Encoder Representations from Transformers (BERT) show promise, standard Cross-Entropy (CE) Loss is biased toward majority classes, and Focal Loss can cause optimization instability when applied early in training. This thesis proposes a framework combining two-stage fine-tuning with Large Language Model (LLM)-based semantic augmentation. The two-stage approach first uses CE Loss to establish stable decision boundaries, then refines them with Focal Loss to emphasize minority classes, avoiding instability while mitigating imbalance. LLM-based augmentation generates semantically refined descriptions through carefully designed prompts, introducing diversity while preserving classification-relevant information. Experimental evaluations on a bilingual dataset of 20,278 English and Vietnamese trademark descriptions demonstrate that the proposed framework significantly outperforms the baseline BERT-based encoder with Support Vector Machine (SVM) approach, which achieves 71.9% accuracy and 69.3% macro-F1. End-to-end fine-tuning of the BERT encoder with CE Loss achieves 79.7% accuracy and 79.1% macro-F1, while the two-stage fine-tuning strategy further improves performance to 81.1% accuracy and 80.2% macro-F1. With LLM-based augmentation, the framework achieves the best performance of 83.6% accuracy and 82.5% macro-F1, validating its effectiveness for imbalanced multilingual multi-class text classification.

Student

(Signature and full name)

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LIST OF ABBREVIATIONS

Abriviation	Full Expression
API	Application Programming Interface
BERT	Bidirectional Encoder Representations from Trans- formers
CE	Cross-Entropy
EDA	Exploratory Data Analysis
GPU	Graphics Processing Unit
LLM	Large Language Model
NLP	Natural Language Processing
OCR	Optical Character Recognition
PCA	Principal Component Analysis
SVM	Support Vector Machine
TF-IDF	Term Frequency-Inverse Document Frequency
UMAP	Uniform Manifold Approximation and Projection

CHAPTER 1. INTRODUCTION

1.1 Motivation

Trademark registration is a critical process in protecting intellectual property rights for businesses and individuals worldwide. The NICE Classification, an international system established under the NICE Agreement of 1957, provides a standardized framework for categorizing goods and services into 45 distinct classes. Accurate classification of trademark applications according to this system is essential for ensuring proper legal protection, avoiding conflicts between similar marks, and maintaining the integrity of trademark registries.

In Vietnam, the Intellectual Property Office processes thousands of trademark applications annually [1], each requiring manual classification by trained examiners. This process is time-consuming, labor-intensive, and susceptible to human error, particularly given the complexity of distinguishing between semantically similar product and service categories. The challenge is further compounded by the bilingual nature of trademark descriptions, which may be submitted in either Vietnamese or English, requiring examiners to possess proficiency in both languages.

The automation of trademark classification through machine learning offers significant potential benefits. An accurate and efficient classification system could substantially reduce the workload of intellectual property offices, accelerate the trademark registration process, and improve consistency in classification decisions. Furthermore, such a system could serve as a valuable tool for trademark applicants, legal professionals, and businesses seeking to understand how their goods and services would be categorized under the NICE Classification framework.

However, developing an effective automated classification system for trademarks presents several technical challenges. The 45-class classification problem involves a large number of categories with varying degrees of semantic overlap. In addition, the distribution of trademark descriptions across classes is highly imbalanced, with some categories containing thousands of samples while others are represented by only a limited number of examples. This imbalance can bias standard machine learning models toward majority classes and degrade performance on minority categories. Moreover, trademark descriptions often include domain-specific terminology, abbreviations, and stylistic variations that complicate semantic interpretation.

1.2 Objectives and Scope of the Graduation Thesis

Recent advances in Natural Language Processing (NLP), particularly the development of pretrained language models such as Bidirectional Encoder Representations from Transformers (BERT) [2], have demonstrated strong performance in text classification tasks. These models learn contextualized representations from large-scale corpora and can be fine-tuned for a wide range of downstream applications. However, their effectiveness in the specific setting of severely imbalanced, bilingual trademark text classification remains an open research problem.

Existing approaches to imbalanced text classification typically rely on techniques such as oversampling minority classes, undersampling majority classes, or modifying the loss function to assign greater importance to underrepresented categories. Focal Loss [3], originally proposed for object detection, has been adapted to text classification to mitigate class imbalance by down-weighting easy-to-classify samples. In addition, Supervised Contrastive Learning [4] has been explored as a means of learning more discriminative representations in the embedding space.

Despite these advances, limitations persist when these methods are applied to domain-specific and highly imbalanced classification tasks. Traditional text data augmentation techniques, such as synonym replacement or back-translation, often fail to preserve domain-specific semantics and may introduce noise that negatively affects classification performance.

Based on these observations, this thesis aims to develop a comprehensive framework for bilingual trademark text classification that explicitly addresses severe class imbalance and domain-specific language. The specific objectives of this research are as follows. First, to establish baseline performance using both traditional machine learning methods and pretrained multilingual language models. Second, to investigate the effectiveness of different loss functions and fine-tuning strategies for handling class imbalance. Third, to propose a novel two-stage fine-tuning strategy that progressively combines Cross-Entropy (CE) Loss and Focal Loss. Fourth, to develop a semantic data augmentation approach based on LLM to improve classification performance, particularly for minority classes.

1.3 Proposed Framework

This thesis adopts an iterative experimental methodology in which each research stage builds upon the findings of the preceding phase. The proposed solution integrates advances in pretrained language models, imbalance-aware training strategies, and generative data augmentation.

The technical approach begins with the selection of a suitable pretrained multilingual encoder. After systematic evaluation of multiple candidates, a pretrained BERT-based model released via Hugging Face is selected as the backbone for its effectiveness in capturing semantic information from bilingual trademark descriptions. The model architecture uses the contextual representation of the [CLS] token as a sentence-level embedding, which is subsequently fed into a linear classification head for multi-class prediction.

To address the class imbalance issue, this thesis adopts a two-stage fine-tuning strategy inspired by the work of Luo et al. [5]. In the first stage, the model is trained using class-weighted CE Loss to establish stable decision boundaries and enable the encoder to adapt to domain-specific characteristics. In the second stage, Focal Loss is applied to further refine these boundaries by emphasizing hard-to-classify samples and underrepresented classes. This staged training strategy helps mitigate optimization instability that may occur when imbalance-aware loss functions are applied from the beginning of training.

To further enhance performance on minority classes, a semantic data augmentation approach based on LLM is introduced. Instead of relying on surface-level transformations, this method employs carefully designed prompts to generate semantically refined trademark descriptions that preserve class-relevant information while increasing lexical diversity. The augmented data effectively expands the training set and provides additional informative examples for underrepresented categories.

The main contributions of this thesis are twofold. First, a two-stage fine-tuning strategy inspired by prior work is adapted and systematically applied to the task of bilingual trademark text classification, demonstrating its effectiveness in addressing severe class imbalance in this domain. Second, a semantic data augmentation approach based on LLM is developed to enhance training data quality and improve performance on underrepresented classes.

1.4 Thesis Organization

The remainder of this thesis is organized as follows. Chapter 2 presents the theoretical background on multi-class text classification, Transformer architecture, BERT, and techniques for handling class imbalance. Chapter 3 describes the methodology, including loss function formulations and the experimental pipeline. Chapter 4 reports the experimental setup, evaluation metrics, data preparation and analysis, and evaluation results across baseline experiments, loss function analysis, and the developed training strategies. Chapter 5 discusses the key contributions in de-

tail, including the adaptation of two-stage fine-tuning strategy and LLM-based data augmentation approach. Finally, Chapter 6 concludes with a summary of findings, limitations, and directions for future research.

CHAPTER 2. RELATED WORK

This chapter introduces the theoretical concepts that underpin the study, including multi-class text classification, text representation methods, Transformer-based language models, and techniques for handling class imbalance. It also provides an overview of related research works, positioning the contributions of this thesis within the current academic landscape while highlighting the novelty and relevance of the adopted methods.

2.1 Multi-class Text Classification Problem

Text classification is one of the fundamental tasks in Natural Language Processing (NLP), where the objective is to assign one or more labels to an input text based on its content. In this thesis, the problem is formulated as a multi-class classification task, in which each text instance belongs to exactly one of N classes, with N being relatively large (45 classes in the NICE trademark classification domain [6]).

Compared to binary classification or small-scale multi-class classification, problems with a large number of classes introduce several significant challenges. First, there is a high degree of semantic overlap among classes, making it difficult to distinguish between closely related categories. In the trademark domain, for example, classes such as “cleaning preparations” and “polishing preparations” share substantial semantic content, requiring fine-grained discrimination capabilities. Second, the dataset often exhibits severe class imbalance, where some classes contain thousands of training samples while others have only a limited number of instances. This imbalance causes standard learning algorithms to be biased toward majority classes, resulting in poor predictive performance for underrepresented categories. Third, learning clear and well-separated decision boundaries in the representation space becomes increasingly difficult as the number of classes grows, since the model must partition a high-dimensional space into many distinct regions.

These challenges cause the performance of traditional classification approaches to degrade substantially when scaling to a large number of classes. Addressing these issues requires careful consideration of both the representation learning strategy and the training objective.

2.2 Text Representation

Text representation plays a crucial role in determining the effectiveness of text classification models. The quality of the input representation directly impacts the classifier’s ability to distinguish between different categories.

Traditional approaches such as Bag-of-Words (BoW) and Term Frequency–Inverse Document Frequency (TF-IDF) represent documents as sparse vectors based on word occurrence statistics. In the BoW model, each document is represented as a vector of word counts, ignoring word order and context. TF-IDF extends this approach by weighting terms according to their frequency within a document relative to their frequency across the corpus, thereby emphasizing discriminative terms while down-weighting common words. Although these methods are simple, computationally efficient, and effective in certain scenarios, they suffer from several fundamental limitations. Specifically, they fail to capture contextual information, as each word is represented independently regardless of its surrounding context. They are also unable to model semantic relationships between words, treating synonyms and related terms as completely distinct features.

Word embeddings, such as Word2Vec [7] and GloVe [8], addressed some of these limitations by learning dense vector representations that capture semantic relationships between words. However, these approaches generate static embeddings that assign the same vector to a word regardless of its context, failing to account for polysemy and context-dependent meaning.

The emergence of contextualized embeddings represents a paradigm shift in text representation. Models such as BERT [2] generate embeddings that depend on the entire sentence context, producing different representations for the same word in different contexts. This development has led to substantial performance improvements across a wide range of NLP tasks and forms the foundation for the approach adopted in this thesis.

2.3 Bidirectional Encoder Representations from Transformers (BERT)

The Transformer architecture, introduced by Vaswani et al. [9], revolutionized natural language processing by demonstrating that attention mechanisms alone, without recurrence or convolution, could achieve state-of-the-art performance. The architecture relies on the self-attention mechanism, which computes relationships between all pairs of positions in a sequence, enabling effective modeling of long-range dependencies. The Transformer encoder consists of multiple stacked layers, each comprising a multi-head self-attention mechanism and a position-wise feed-forward network, with residual connections and layer normalization to facilitate training of deep networks. Since self-attention is permutation-invariant, positional encodings are added to input embeddings to inject positional information.

Building upon the Transformer architecture, BERT [2] is a deep bidirectional Transformer encoder designed to learn rich contextual representations by jointly

considering both left and right contexts at every layer. The pretrained BERT model can be adapted for various downstream tasks through fine-tuning. For text classification, the contextual representation of the special `[CLS]` token from the final encoder layer is commonly used as a global representation of the input sequence.

2.4 Fine-tuning Strategies for Pretrained Language Models

Fine-tuning refers to the process of adapting a pretrained model by updating its parameters for a specific downstream task. The standard approach involves initializing the model with pretrained weights and then training on task-specific data using a task-appropriate loss function. However, several considerations influence the effectiveness of fine-tuning.

The choice of which layers to update during fine-tuning significantly impacts both performance and training stability. Full fine-tuning updates all parameters of the pretrained encoder along with the task-specific layers. While this approach allows maximum adaptation to the target task, it risks overfitting when labeled data is limited and may cause catastrophic forgetting of the general linguistic knowledge encoded during pretraining.

Partial fine-tuning, also known as gradual unfreezing, updates only a subset of layers during training. Studies have shown that lower layers of BERT encode general linguistic features such as syntax and morphology, while upper layers capture more task-specific and semantic information[10]. Consequently, fine-tuning only the upper layers offers multiple advantages. First, it reduces the risk of overfitting, especially when the labeled dataset is limited. Second, it leads to a more stable training process by preventing excessive updates to lower layers that encode general linguistic knowledge. Third, this strategy helps preserve the rich and transferable language representations learned during the pretraining phase while still allowing task-specific adaptation.

The learning rate and its schedule also play critical roles in fine-tuning. Lower learning rates are typically used for pretrained layers to avoid disrupting learned representations, while higher learning rates may be applied to randomly initialized task-specific layers. Learning rate warmup and decay schedules help stabilize training and improve final performance.

2.5 Class Imbalance and Loss Functions

Data imbalance is a pervasive challenge in real-world classification problems, where the number of samples varies significantly across different classes. In such scenarios, standard learning algorithms tend to be biased toward majority classes, as these classes dominate the training objective. This bias manifests as high accu-

racy on frequent categories but poor predictive performance for minority classes, which may be equally or more important for the application.

Cross-Entropy (CE) Loss is the standard objective function for multi-class classification. It measures the dissimilarity between the predicted probability distribution and the true label distribution, penalizing incorrect predictions proportionally to the model’s confidence. However, CE Loss treats all samples equally regardless of their class frequency, leading to optimization dominated by majority classes in imbalanced settings.

Several approaches have been proposed to address class imbalance. Class-weighted CE Loss modifies the loss function to assign higher weights to minority classes, increasing their contribution to the gradient updates. A common balanced weighting scheme computes class weights as $w_c = n_{\text{samples}} / (n_{\text{classes}} \times n_c)$, where n_c is the number of samples in class c , and these weights are often normalized to have unit mean for training stability. However, determining appropriate weights can be challenging, and aggressive weighting may lead to training instability.

Focal Loss [3] was introduced as a more principled approach to handling class imbalance. The key insight is that easy-to-classify samples, which are often from majority classes, dominate the CE loss and contribute little useful gradient signal. Focal Loss addresses this by introducing a modulating factor that down-weights the loss contribution from well-classified samples, allowing the model to focus on hard examples that are more informative for learning. A key component of Focal Loss is the focusing parameter γ , which controls the degree to which easy examples are down-weighted. As the value of γ increases, the loss function places progressively more emphasis on difficult samples, thereby improving robustness to class imbalance.

Supervised Contrastive Learning [4] offers an alternative approach by operating at the representation level rather than the classification level. The objective encourages samples from the same class to be close in the embedding space while pushing samples from different classes apart. This approach can produce more discriminative representations that benefit downstream classification. However, as investigated in this thesis, applying contrastive learning to models that were already pretrained with similar objectives may provide limited additional benefit.

Recent work has explored two-stage training strategies that combine different loss functions [5]. The motivation is that different loss functions may be optimal at different stages of training. For example, CE Loss may be preferable in early training to establish stable representations, while imbalance-aware losses like Focal

Loss may be more beneficial in later stages for refining decision boundaries. This insight motivates the two-stage fine-tuning strategy adopted in this thesis.

2.6 Data Augmentation for Text Classification

Data augmentation is a widely used technique to increase the effective size and diversity of training data, particularly beneficial when labeled examples are scarce. In computer vision, augmentation techniques such as rotation, cropping, and color jittering have become standard practice. However, augmenting text data is more challenging due to the discrete nature of language and the need to preserve semantic meaning.

Back-translation is a popular augmentation technique that translates text to another language and then back to the original language, producing paraphrased versions with different surface forms but similar meaning. However, this approach depends on the quality of the translation system and may not preserve domain-specific terminology accurately.

The emergence of LLM has opened new possibilities for text augmentation. These models can generate coherent and contextually appropriate text conditioned on prompts, enabling more sophisticated augmentation strategies. Rather than making surface-level modifications, LLM can be instructed to rephrase, summarize, or expand text while preserving its semantic content. This capability is particularly valuable for domain-specific applications where maintaining technical accuracy is essential.

In this thesis, an LLM-based augmentation strategy is developed that focuses on semantic refinement rather than simple paraphrasing. The approach uses carefully designed prompts to generate improved descriptions that are more standardized, informative, and classification-friendly, while strictly preserving the original class labels. This method addresses the heterogeneity and ambiguity present in real-world trademark descriptions, producing augmented data that provides clearer semantic signals for the classification model.

CHAPTER 3. METHODOLOGY

Chapter 2 has established the theoretical background and core technologies underpinning this research. Building upon this foundation, the present study aims to enhance the performance of a multi-class text classification system under data imbalance conditions, with a specific focus on trademark-related text classification. An iterative experimental framework is adopted, in which each research stage is informed by and extends the findings of the preceding phase.

3.1 Problem Statement

The core challenge addressed in this research is the automated classification of trademark descriptions into the appropriate NICE classes based solely on textual product and service descriptions. This task requires the model to understand the semantic content of trademark applications and map them to one of 45 distinct NICE classes, which serve as the standardized international framework for categorizing goods and services in trademark registration.

The NICE Classification system, as published by the Intellectual Property Office of Vietnam [6], organizes goods and services into a structured table format. The classification table consists of 45 classes, where classes 1 through 34 cover goods, and classes 35 through 45 cover services. Each class entry contains a class number, a class heading that provides a general description of the category, and an extensive alphabetical list of specific goods or services that fall within that class.

Table 3.1 presents illustrative examples extracted from the NICE Classification table, showing entries from multiple classes. This example demonstrates the granularity and specificity required for accurate classification, as well as the diversity of products and services across different NICE classes. The classification task becomes particularly challenging when dealing with bilingual descriptions, as trademark applications may be submitted in either English or Vietnamese, requiring the model to understand semantic equivalence across languages. Furthermore, the dataset exhibits severe class imbalance, where certain NICE classes contain thousands of training samples while others are represented by only a limited number of examples.

The overall methodology is structured into five main phases. First, an Exploratory Data Analysis (EDA) is conducted to examine the characteristics of the dataset, including the distribution of English and Vietnamese samples, as well as to identify class imbalance issues that negatively affect classification performance. Second,

Table 3.1: Some samples from the NICE Classification table

Class	English	Vietnamese
1	Glass-staining chemicals	Hoá chất nhuộm màu kính
2	Printing ink	Mực in
7	Leather paring machines	Máy cắt xén da thuộc
10	Ophthalmoscopes	Kính soi đáy mắt
41	Providing information in the field of education	Cung cấp thông tin trong lĩnh vực giáo dục
45	Legal services in relation to the negotiation of contracts for others	Dịch vụ pháp lý liên quan đến đàm phán hợp đồng cho người khác

TF-IDF and BERT-based models combined with Support Vector Machine (SVM) classifier is employed to evaluate the embedding models performance using a multilingual dataset. Third, based on the selected direction, the model architecture is further refined by replacing the SVM classifier with a linear classification head, enabling end-to-end fine-tuning. In this phase, different feature extraction strategies, namely the [CLS] token representation and mean pooling, are systematically compared. Fourth, various loss functions are investigated, and a two-stage fine-tuning strategy is adapted to mitigate the adverse effects of data imbalance during training. Finally, data augmentation is performed by leveraging LLM with carefully designed prompts to generate enriched and refined product and service descriptions, thereby improving the overall quality and diversity of the training data.

3.2 Loss Functions

This section presents the mathematical formulations of the loss functions employed in this study. The choice of loss function plays a critical role in guiding the optimization process, particularly under class imbalance conditions.

3.2.1 Cross-Entropy Loss

Cross-Entropy (CE) Loss is the standard objective function for multi-class classification. Given a training sample with true class label y and predicted probability distribution $\hat{p}$ over C classes, the standard CE Loss is defined as:

$$\mathcal{L}_{\text{CE}} = - \sum_{c=1}^C y_c \log(\hat{p}_c) = - \log(\hat{p}_y), \quad (3.1)$$

where y_c is the one-hot encoded label (equal to 1 if $c = y$ and 0 otherwise), and $\hat{p}_y$ denotes the predicted probability for the true class. CE Loss penalizes incorrect predictions proportionally to the confidence of the model, encouraging high

probability assignments to the correct class.

To address class imbalance, class-weighted CE Loss is employed in this study. The class weights are computed using the balanced weighting scheme:

$$w_c = \frac{n_{\text{samples}}}{n_{\text{classes}} \times n_c}, \quad (3.2)$$

where n_{samples} is the total number of training samples, n_{classes} is the number of classes, and n_c is the number of samples belonging to class c . The weights are then normalized such that their mean equals 1:

$$w_c^{\text{norm}} = \frac{w_c}{\bar{w}}, \quad \text{where} \quad \bar{w} = \frac{1}{C} \sum_{c=1}^C w_c. \quad (3.3)$$

The weighted CE Loss is then defined as:

$$\mathcal{L}_{\text{CE}}^{\text{weighted}} = -w_y^{\text{norm}} \log(\hat{p}_y), \quad (3.4)$$

where w_y^{norm} is the normalized weight for the true class y . This weighting scheme assigns higher importance to minority classes, thereby mitigating the bias toward majority classes during training.

3.2.2 Focal Loss

Focal Loss was introduced to address class imbalance by down-weighting easy-to-classify samples and focusing the learning process on hard examples. The formulation extends CE Loss with a modulating factor:

$$\mathcal{L}_{\text{Focal}} = -\alpha_y (1 - \hat{p}_y)^\gamma \log(\hat{p}_y), \quad (3.5)$$

where α_y is a class-dependent weighting factor that balances the importance of different classes, and $\gamma \geq 0$ is a focusing parameter that controls the rate at which easy examples are down-weighted. When $\gamma = 0$, Focal Loss reduces to standard CE Loss. As γ increases, the loss contribution from well-classified samples (high $\hat{p}_y$) diminishes, allowing the model to concentrate on harder and minority-class examples.

3.2.3 Contrastive Loss

Contrastive Loss operates at the representation level by encouraging samples from the same class to be close in the embedding space while pushing samples from different classes apart. Given a batch of samples, let z_i and z_j denote the normalized embeddings of samples i and j , respectively. The contrastive loss for

sample i is defined as:

$$\mathcal{L}_{\text{Contrastive}}^{(i)} = -\frac{1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(z_i \cdot z_p / \tau)}{\sum_{a \in A(i)} \exp(z_i \cdot z_a / \tau)}, \quad (3.6)$$

where $P(i)$ is the set of indices of samples in the batch that share the same class label as sample i (positive pairs), $A(i)$ is the set of all sample indices in the batch excluding i , and $\tau > 0$ is a temperature parameter that controls the sharpness of the similarity distribution. The overall Contrastive Loss is computed by averaging over all samples in the batch.

3.3 Baseline Methods

Baseline experiments were conducted to establish reference performance and to guide the selection of an appropriate modeling approach for the trademark text classification task. First, a traditional machine learning baseline based on Term Frequency–Inverse Document Frequency (TF-IDF) feature extraction combined with an SVM classifier was evaluated. This approach serves as a strong non-neural baseline for comparison with neural language models.

Next, several multilingual BERT-based models [2] are investigated, in which contextual embeddings extracted from the pretrained models are used as input features for an SVM classifier. These experiments aim to assess the capability of multilingual pretrained language models to capture semantic information from trademark descriptions in both English and Vietnamese.

In addition, two feature extraction strategies are systematically compared: the contextual representation of the [CLS] token and mean pooling over all token embeddings. The best-performing model and feature extraction strategy are selected based on experimental results for subsequent fine-tuning experiments.

3.4 Architecture Enhancement

Based on the findings from the baseline experiments, the model architecture is further refined to enable end-to-end optimization. While pretrained encoders provide strong feature representations, the use of frozen embeddings with an SVM limits the model’s ability to adapt its representations to the specific characteristics of the trademark classification task.

To overcome this limitation, end-to-end fine-tuning is adopted by replacing the SVM with a lightweight linear classification head and allowing gradient updates to propagate through the encoder layers. This architectural modification enables the model to jointly optimize both the feature extractor and the classifier, leading to

more task-specific and discriminative representations. The [CLS] token representation is used as the sentence-level embedding based on the baseline comparison.

The linear classification head is defined as a simple affine transformation followed by a softmax activation:

$$\hat{p} = \text{softmax}(W \cdot h_{[\text{CLS}]} + b), \quad (3.7)$$

where $h_{[\text{CLS}]} \in \mathbb{R}^d$ is the contextual embedding of the [CLS] token extracted from the final layer of the encoder, $W \in \mathbb{R}^{C \times d}$ and $b \in \mathbb{R}^C$ are learnable parameters, and $\hat{p} \in \mathbb{R}^C$ is the predicted probability distribution over C classes.

This architectural design allows gradient signals from the classification loss to be propagated through the entire network, enabling the pretrained language model to be effectively fine-tuned for the domain-specific and imbalanced nature of the dataset.

3.5 Loss Function Experiments

A set of loss functions was evaluated to investigate their effectiveness in handling the class imbalance inherent in the trademark text classification task. Specifically, class-weighted CE Loss, Focal Loss [3], and Contrastive Loss [4] were considered, with their mathematical formulations presented in Section 3.1.

Class-weighted CE Loss serves as a commonly used baseline for multi-class classification. Focal Loss was introduced to address class imbalance by down-weighting easy-to-classify samples and focusing the learning process on hard and minority-class examples. In contrast, Contrastive Loss operates at the representation level by encouraging samples from the same class to be close in the embedding space while pushing samples from different classes apart.

3.6 Two-Stage Fine-tuning Strategy for Imbalanced Data

To address the severe class imbalance identified during the exploratory data analysis, a two-stage fine-tuning strategy inspired by the work of Luo et al. [5] was adapted and evaluated. The primary objective of this strategy is to stabilize model optimization while gradually introducing imbalance-aware learning signals.

In the first stage, the model is fine-tuned using class-weighted CE Loss. This stage focuses on enabling the linear classification head to learn reliable decision boundaries while accounting for class imbalance through normalized class weights, thereby providing a stable initialization for subsequent training. The optimization

objective in this stage is:

$$\mathcal{L}_{\text{Stage 1}} = \mathcal{L}_{\text{CE}}^{\text{weighted}} = -w_y^{\text{norm}} \log(\hat{p}_y), \quad (3.8)$$

where w_y^{norm} is the normalized class weight computed using the balanced weighting scheme described in Section 3.2.

In the second stage, Focal Loss is applied to explicitly address class imbalance by reducing the contribution of easy-to-classify majority-class samples and emphasizing harder and minority-class examples. This stage encourages the model to refine its representations and decision boundaries in a manner that improves performance on underrepresented classes. The optimization objective transitions to:

$$\mathcal{L}_{\text{Stage 2}} = \mathcal{L}_{\text{Focal}} = -\alpha_y (1 - \hat{p}_y)^\gamma \log(\hat{p}_y). \quad (3.9)$$

This progressive training framework is designed to combine the stability of CE Loss with the imbalance-awareness of Focal Loss, aiming to improve classification performance on underrepresented classes without introducing optimization instability.

3.7 Data Augmentation with LLM

To further enhance model robustness and alleviate data scarcity in minority classes, a data augmentation strategy based on LLM was employed. Specifically, LLM was leveraged to generate improved and semantically enriched trademark descriptions through carefully designed prompting techniques.

A structured prompt template was constructed to guide the LLM in rewriting original product and service descriptions while preserving their semantic content and class labels. The prompt template used for data augmentation is shown below:

You are an expert in NICE Classification for trademarks.
Your task is to provide a concise definition of the product/service based on the description below.

Requirements:

- Describe its nature (what it is/its properties) and purpose (primary function/utility)
- The output should be in {language} language only (strictly enforced)
- Maximum 50 words (strictly enforced)
- Use scientific and professional tone
- Do not include any redundant prefix or suffix in the

output

Description: {description}

Output:

The prompt explicitly instructs the model to enhance clarity, completeness, and domain relevance, ensuring that the generated descriptions remain consistent with the corresponding NICE Classification categories. This approach allows the augmented samples to introduce lexical and syntactic diversity without altering the underlying semantic intent.

Table 3.2 presents example inputs and their corresponding LLM-generated outputs, demonstrating how the prompt template transforms original descriptions into refined, standardized versions that maintain semantic consistency while improving clarity and information density.

Table 3.2: Example inputs and outputs from LLM-based description refinement

Input	Output
Camera quan sát phía sau dùng cho xe cộ	Thiết bị giám sát hành vi giao thông, cung cấp hình ảnh trực tiếp phía sau xe cơ giới. Dùng để hỗ trợ lái xe, tăng cường an toàn và giảm thiểu tai nạn.
Rearview cameras for vehicles	Rearview cameras are electronic devices providing a visual representation of the area behind a vehicle. Their purpose is to enhance driver safety and facilitate maneuvering by improving visibility during reversing and parking operations.

The prompt template enforces several constraints to ensure high-quality output. First, the nature and purpose of the product or service must be clearly described. Second, the output language must match the original input language. Third, the output length is limited to ensure conciseness and information density. Fourth, a scientific and professional tone is required to maintain consistency with domain-specific terminology.

The augmented data are incorporated into the training set and used in conjunction with the adapted two-stage fine-tuning strategy. This combined approach aims to address both the data scarcity in minority classes and the optimization challenges posed by severe class imbalance.

CHAPTER 4. EXPERIMENT AND EVALUATION

Based on the methodology proposed in Chapter 3, this chapter presents the systematic experimental process conducted to develop and optimize the trademark classification system. The experiments are designed to answer the defined research questions through an incremental and controlled evaluation strategy. The experimental pipeline begins with a baseline approach that employs pre-trained BERT embeddings combined with an SVM classifier. Subsequently, the architecture is progressively improved by replacing the SVM with a linear classifier head to enable end-to-end tuning, investigating alternative loss functions to mitigate data imbalance, introducing a two-stage fine-tuning strategy, and finally augmenting the training data using LLM-generated trademark descriptions.

4.1 Experimental Setup

4.1.1 Hardware and Software Environment

All experiments were conducted in a controlled computational environment to ensure reproducibility. The experimental setup utilized an NVIDIA RTX 4090 Graphics Processing Unit (GPU), providing high computational throughput for large-scale neural network training. The operating system was Ubuntu 22.04, a stable Linux-based distribution widely used in research environments. All experiments were executed on cloud-based infrastructure provided by Vast.ai. The models were implemented using the PyTorch deep learning framework in conjunction with the Hugging Face Transformers library, and all experimental pipelines were developed and executed using the Python programming language.

4.1.2 Evaluation Metrics

Model performance is evaluated using standard multi-class classification metrics. Accuracy measures the overall proportion of correctly classified samples across all classes and is defined as:

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of samples}}. \quad (4.1)$$

Precision and recall are first defined at the class level. Precision measures the correctness of the model's positive predictions for a given class, while recall evaluates the model's ability to retrieve all relevant samples belonging to that class. These metrics are defined as follows:

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}, \quad (4.2)$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c}, \quad (4.3)$$

where TP_c , FP_c , and FN_c denote the number of true positives, false positives, and false negatives for class c , respectively.

The F1-score is defined as the harmonic mean of precision and recall, providing a balanced measure of classification performance:

$$F1_c = \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}. \quad (4.4)$$

To account for class imbalance in multi-class settings, both macro-averaged and weighted F1-scores are reported. The macro-averaged F1-score assigns equal importance to all classes and is computed as:

$$\text{Macro-F1} = \frac{1}{C} \sum_{c=1}^C F1_c, \quad (4.5)$$

where C denotes the total number of classes.

In contrast, the weighted F1-score aggregates class-wise F1-scores according to their relative frequencies in the dataset:

$$\text{Weighted-F1} = \sum_{c=1}^C w_c \cdot F1_c, \quad (4.6)$$

where

$$w_c = \frac{N_c}{\sum_{j=1}^C N_j}, \quad (4.7)$$

and N_c denotes the number of samples belonging to class c , while the denominator represents the total number of samples across all C classes.

While weighted F1 reflects overall performance dominated by majority classes, macro-F1 is particularly sensitive to performance on minority classes. Therefore, both metrics are reported to provide a comprehensive evaluation under severe class imbalance conditions.

4.1.3 Data Preparation and Analysis

The source data were obtained from an official PDF document published by the Intellectual Property Office of Vietnam [6]. The document was processed using the Optical Character Recognition (OCR) via the Application Programming Interface (API) provided by Mistral AI to extract the raw textual content. The OCR output

was subsequently post-processed to remove noise and to extract structured information from the classification tables, which was then used to construct the dataset for the trademark text classification task.

The resulting dataset consists of trademark descriptions written in both English and Vietnamese, corresponding to multiple categories defined under the NICE Classification system. Each data instance is associated with a single class label, resulting in a large-scale multi-class classification problem comprising 45 distinct classes.

The bilingual nature of the dataset introduces additional linguistic diversity, while the domain-specific terminology inherent in trademark descriptions poses further challenges for semantic representation and accurate classification.

An EDA was conducted to examine the distribution of samples across different trademark classes. The purpose of this analysis is to identify potential class imbalance issues, which are expected to bias the learning process toward majority classes and degrade the model’s performance on minority classes. Identifying such issues at an early stage provides a critical foundation for justifying subsequent architectural and training decisions.

Figure 4.1 presents the class-wise sample distribution over all 45 classes. The bar chart reveals a highly skewed distribution, in which a small subset of classes contains a disproportionately large number of samples, while many classes are represented by only a limited number of instances. This observation is further supported by the accompanying box plot, which highlights substantial variance in class sizes as well as the presence of extreme outliers corresponding to majority classes.

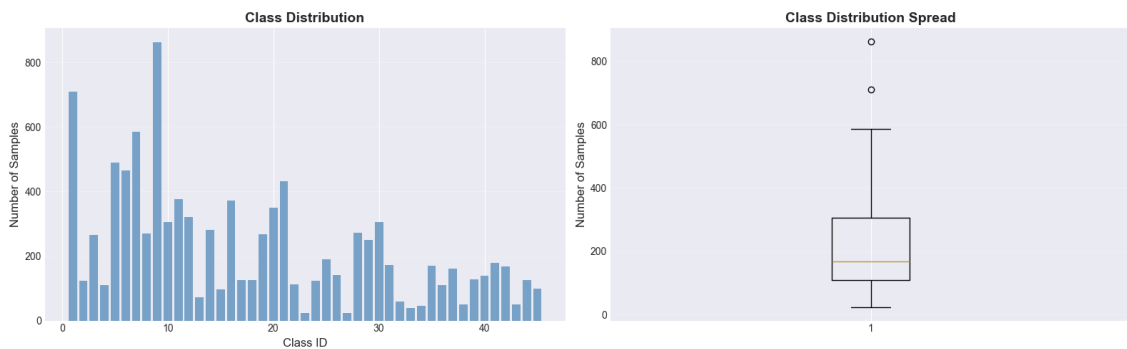


Figure 4.1: Class distribution across 45 trademark classes.

Such a severe class imbalance is expected to bias the learning process toward majority classes, thereby degrading classification performance on underrepresented categories.

In addition to class-level imbalance, the distribution of samples across languages

was also analyzed. As summarized in Table 4.1, the dataset contains an equal number of English and Vietnamese samples, resulting in a balanced bilingual composition. This balance ensures that performance differences observed in later experiments are not primarily driven by language dominance, thereby enabling a fair evaluation of multilingual representation learning methods.

Table 4.1: Dataset statistics by language

Language	Number of Samples
English	10,139
Vietnamese	10,139
Combined	20,278

Following the exploratory analysis, the dataset was partitioned into training, validation, and test sets using a fixed split ratio of 80%, 10%, and 10%, respectively. The split was performed in a stratified manner with respect to class labels to ensure that all 45 classes are consistently represented in each subset. This strategy prevents class omission in the validation or test sets and allows reliable performance evaluation under severe class imbalance conditions. The resulting data partitions are used consistently across all subsequent experiments to ensure fair and reproducible comparisons.

4.2 Baseline Experiments

Baseline experiments were conducted to systematically evaluate the impact of feature representation and pooling strategy on the trademark text classification task. To ensure a fair and controlled comparison across all configurations, an SVM classifier was employed consistently in all baseline experiments.

As a traditional baseline, a TF-IDF representation was used in combination with an SVM. The TF-IDF features were constructed using unigram and bigram representations with a fixed vocabulary size of 5,000. Class-balanced weights were applied during training to partially alleviate the effects of data imbalance.

In addition to TF-IDF features, several pretrained multilingual BERT-based encoders were evaluated. For each encoder, sentence-level representations were extracted using two different pooling strategies: the contextual embedding of the [CLS] token and mean pooling over all token embeddings. These representations were then used as input features for the SVM classifier, allowing performance differences to be attributed solely to representation quality.

Evaluation on a bilingual dataset captures realistic trademark classification settings, where product and service descriptions frequently span multiple languages,

while a unified experimental setup guarantees consistency across all models and pooling strategies.

To ensure architectural comparability, all evaluated BERT-based encoders share a similar model configuration, consisting of approximately 300 million parameters, 12 Transformer layers, and a hidden embedding size of 768. This design choice ensures that observed performance differences primarily stem from representational effectiveness rather than model scale.

Table 4.2 presents the baseline experimental results obtained with different feature representations and pooling strategies. Performance is reported in terms of accuracy, macro-averaged F1 score, and weighted F1 score to account for the imbalanced nature of the dataset.

For brevity, the following abbreviations are used: MPNet denotes *paraphrase-multilingual-mpnet-base-v2*, and E5 denotes *multilingual-e5-base*. All pretrained BERT-based encoders evaluated in the baseline experiments are publicly available models released on the Hugging Face platform.

Table 4.2: Baseline results with SVM classifier.

Encoder	Pooling	Accuracy	Macro-F1	Weighted-F1
TF-IDF	–	62.0%	60.5%	64.9%
MPNet	CLS	71.9%	69.3%	72.1%
MPNet	Mean Pooling	71.2%	68.8%	71.2%
E5	CLS	69.7%	68.8%	69.8%
E5	Mean Pooling	69.4%	68.3%	69.4%

The experimental results demonstrate that pretrained multilingual BERT-based encoders substantially outperform the traditional TF-IDF baseline in the bilingual trademark classification setting. This confirms the effectiveness of contextualized semantic representations for capturing nuanced product and service descriptions across languages.

Across all evaluated encoders, the [CLS] pooling strategy consistently yields superior performance compared to mean pooling, indicating that the [CLS] token provides a more discriminative sentence-level representation for this task.

Among the evaluated models, the BERT-based encoder *paraphrase-multilingual-mpnet-base-v2* achieves the best overall performance across all evaluation metrics. Consequently, this encoder with [CLS] pooling is selected as the backbone model for subsequent end-to-end fine-tuning and advanced training strategies.

4.3 Loss Function Experiments

4.3.1 Experiments

While the baseline experiments demonstrated that pretrained encoders provide strong feature representations, the use of frozen embeddings with an SVM classifier limits the model’s ability to adapt its representations to the specific characteristics of the trademark classification task. To overcome this limitation, end-to-end fine-tuning was adopted, replacing the SVM with a lightweight linear classification head and allowing gradient updates to propagate through the encoder layers. This enables the model to jointly optimize both the feature extractor and the classifier, potentially leading to more task-specific and discriminative representations.

Building upon the baseline findings, additional experiments were conducted to investigate the impact of different loss functions and fine-tuning depths on classification performance. All experiments in this section utilized the selected BERT-based backbone model, combined with a linear classification head and sentence representations derived from the [CLS] token.

To analyze the interaction between loss functions and parameter updating strategies, three unfreezing configurations were considered. In the first configuration, only the linear classification head was updated, while the encoder remained fully frozen. In the second configuration, the linear head and the last Transformer layer of the encoder were unfrozen. In the third configuration, the linear head and the last four Transformer layers were unfrozen, allowing for deeper task-specific adaptation while preserving the stability of lower-level representations.

Under each unfreezing configuration, four loss function settings were evaluated. First, the model was trained using Contrastive Loss alone to assess whether additional representation-level supervision could improve task performance. Second, a hybrid objective combining 30% Contrastive Loss and 70% class-weighted CE Loss was explored to balance representation learning with discriminative classification. Third, class-weighted CE Loss was employed as a strong baseline for multi-class classification. Finally, Focal Loss was applied to explicitly address severe class imbalance by emphasizing hard-to-classify and minority-class samples.

Table 4.3 summarizes the experimental results across different loss functions and unfreezing strategies. Performance is reported in terms of accuracy, macro-averaged F1 score, and weighted F1 score.

The results consistently indicate that training with Contrastive Loss alone leads to a degradation in classification performance across all unfreezing configurations.

Table 4.3: Loss functions comparison across unfreezing strategies.

Loss Function	Unfrozen BERT Layers	Accuracy	Macro-F1	Weighted-F1
Contrastive	Last 1 layer	67.4%	64.6%	67.5%
Contrastive	Last 4 layers	78.1%	75.1%	78.1%
Hybrid	Last 1 layer	68.0%	65.9%	68.1%
Hybrid	Last 4 layers	79.5%	77.9%	79.5%
CE	Last 1 layer	68.2%	66.1%	68.3%
CE	Last 4 layers	79.7%	79.1%	79.7%
Focal	Last 1 layer	69.3%	67.0%	69.5%
Focal	Last 4 layers	78.5%	77.9%	78.5%

This is likely attributable to the fact that the backbone model was already pre-trained using contrastive learning objectives; thus, continued contrastive training provided limited additional representational benefit and may have interfered with task-specific adaptation.

The hybrid loss configuration, which combines Contrastive Loss and class-weighted CE Loss, yields moderate improvements compared to pure contrastive training, particularly when deeper encoder layers are unfrozen. However, this hybrid objective remains inferior to both class-weighted CE Loss and Focal Loss in terms of overall accuracy and macro-averaged F1 score.

Among all evaluated loss functions, class-weighted CE Loss achieves the best overall performance across all metrics. Focal Loss ranks second, providing competitive results but failing to surpass class-weighted CE Loss.

Based on these findings, loss functions involving Contrastive Loss were excluded from subsequent experiments. The remaining training strategies focus on class-weighted CE Loss and Focal Loss, as they demonstrated superior effectiveness for the trademark text classification task under data imbalance conditions.

4.3.2 Learning Curve Analysis

To understand the training dynamics and convergence behavior across different loss functions and architectural configurations, the learning curves for all experimental settings were examined. Figure 4.2 presents the training and validation performance trajectories over 10 epochs for each loss function, comparing models with 1-layer and 4-layer unfreezing strategies.

The learning curves show that models with 4-layer unfreezing always do better than 1-layer models for all loss functions. In training, 4-layer models reach accuracies above 0.90, while 1-layer models stop improving around 0.68-0.70. Validation results follow the same pattern, with 4-layer models doing better than 1-layer mod-

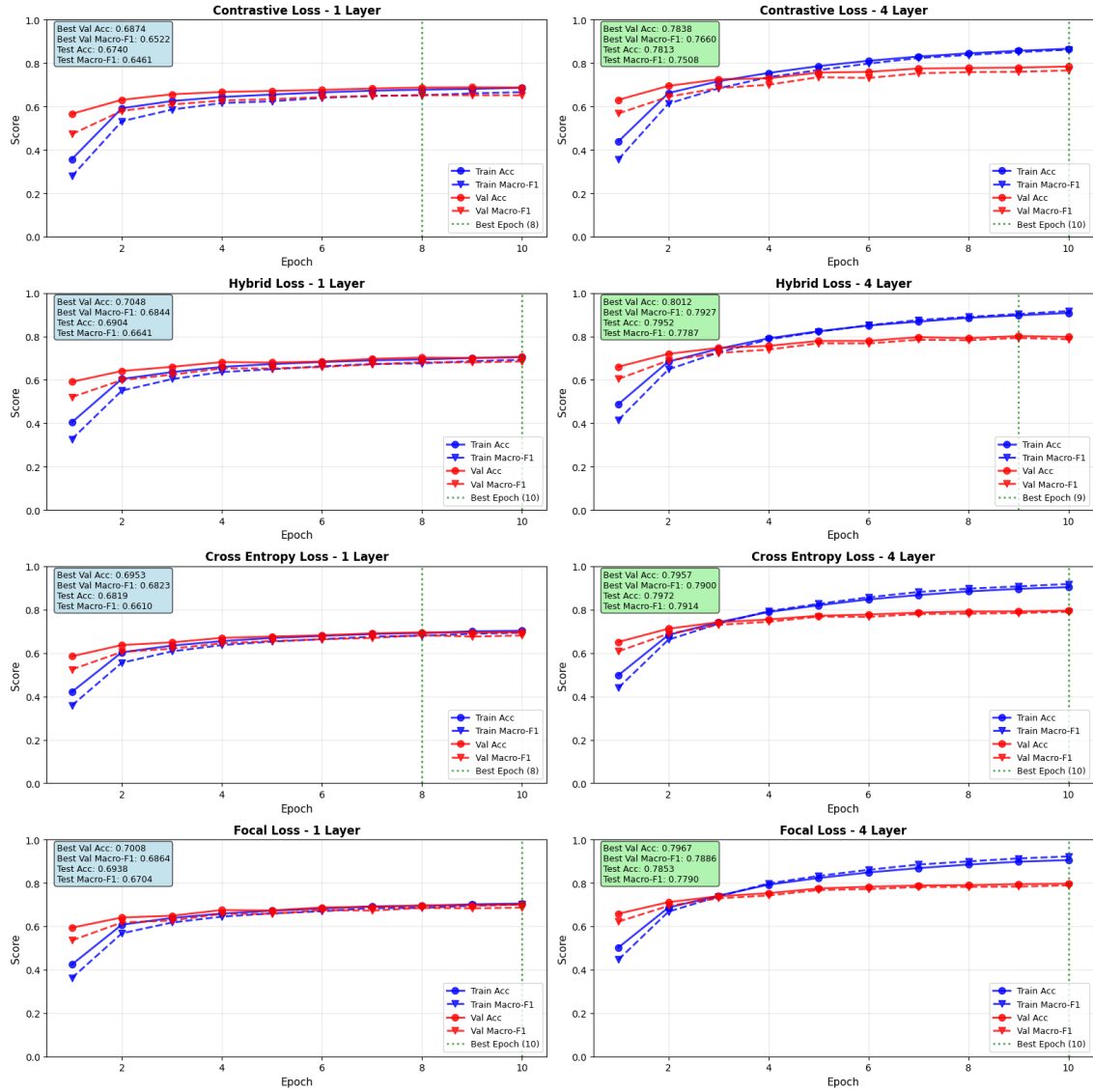


Figure 4.2: Learning curves comparing different loss functions and unfreezing strategies.

els by about 0.09-0.10 in accuracy and 0.10-0.11 in macro-F1.

4.3.3 Representation and Error Analysis

To gain deeper insights into the behavior of the best-performing model, an analysis was conducted on the test set using the configuration with the BERT-based encoder, a linear classification head, [CLS] token representations, class-weighted CE Loss, and unfreezing of the last four encoder layers. This analysis focuses on the geometric structure of learned representations and error patterns across classes.

First, high-dimensional sentence embeddings extracted from the test set were projected into a lower-dimensional space. Principal Component Analysis (PCA) was applied to reduce the embedding dimensionality to 64, preserving 94.12% of the total variance, indicating that the semantic structure of the data remains largely intact. Subsequently, Uniform Manifold Approximation and Projection (UMAP) was employed to further reduce the representations to two dimensions for visualization. Figure 4.3 illustrates the resulting UMAP projection, where each point represents a test sample and class centroids are highlighted.

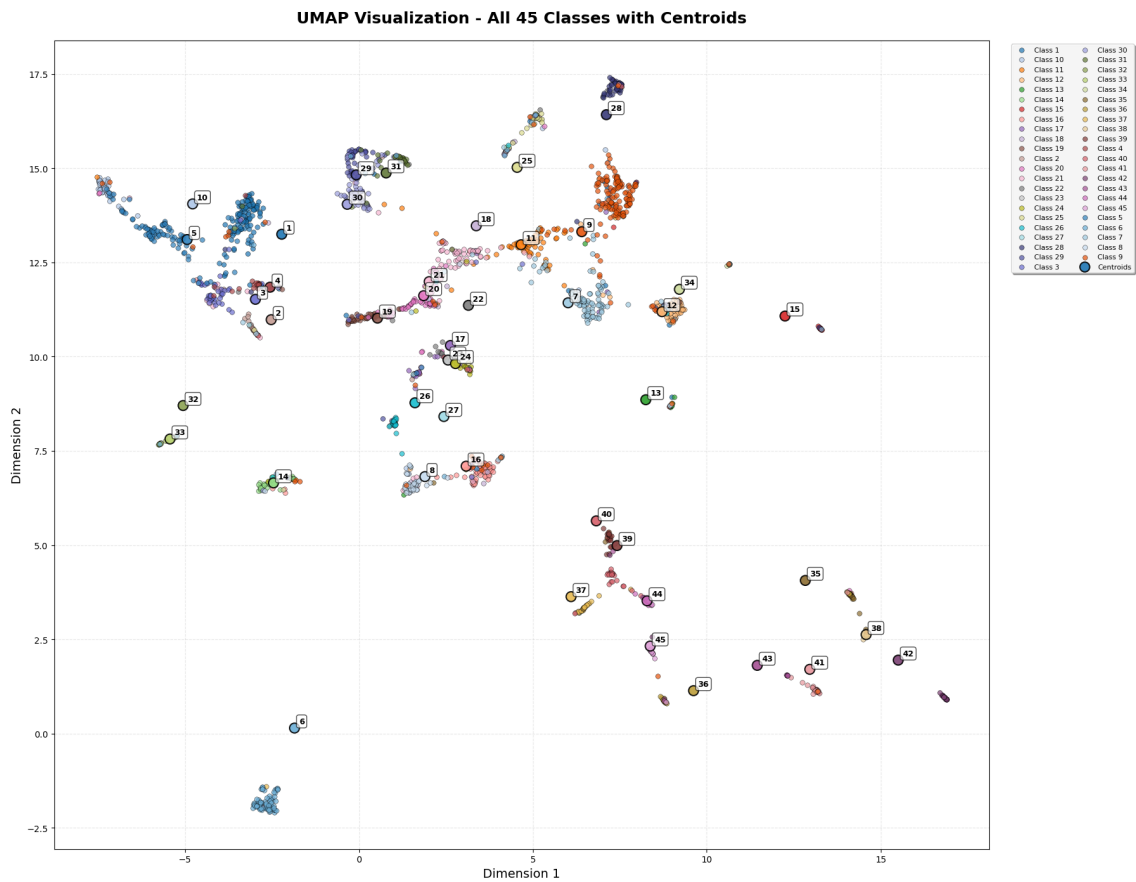


Figure 4.3: UMAP visualization of test set embeddings with class-weighted CE Loss.

The visualization reveals that while several classes form compact and well-separated clusters, others exhibit significant overlap. Notably, sixteen classes ex-

hibit at least one metric (precision, recall, or F1) below the threshold of 0.7, including Classes 2, 3, 4, 13, 17, 18, 20, 21, 22, 23, 24, 26, 30, 33, 34, and 40. These classes tend to be located in densely populated regions or near the boundaries of larger clusters, typically corresponding to categories with relatively few training samples and highlighting the persistent impact of class imbalance.

Despite the use of strategies to address class imbalance, the model remains biased toward majority classes. This behavior is particularly evident in minority classes that exhibit high precision but low recall; while the model’s predictions for these classes are often correct, it fails to retrieve a substantial portion of their true instances. Class 40 serves as a representative example with a precision of 0.85 but a recall of only 0.61, where conservative decision boundaries favor correctness over coverage. Similarly, Class 20 exhibits a precision of 0.77 but a recall of only 0.64.

Additionally, several class centroids lie in close proximity, forming dense semantic regions. While dimensionality reduction techniques like UMAP can introduce minor distortions, the consistent misclassification patterns observed in these regions suggest that the overlap arises primarily from genuine semantic similarity between trademark descriptions rather than visualization artifacts.

To further investigate misclassification behavior, the confusion matrix of the same model configuration is presented in Figure 4.4. For clarity, diagonal elements corresponding to correct predictions are suppressed, allowing the visualization to emphasize inter-class misclassification patterns rather than dominant correct classifications.

The off-diagonal confusion matrix reveals that misclassifications frequently occur between classes whose centroids are close in the UMAP space, indicating that these errors are systematic rather than random. By focusing on the off-diagonal elements, the analysis highlights error structures induced by semantic similarity and data imbalance that would otherwise be obscured by the large number of correct predictions.

Moreover, major classes with a large number of training samples—such as Classes 1, 5, 7, and 9—are frequently predicted for samples belonging to minority classes. This observation further confirms that data imbalance causes decision boundaries to be skewed toward dominant categories, leading to reduced recall for underrepresented classes.

Overall, the analysis demonstrates that although the fine-tuned class-weighted CE-based model achieves strong overall performance, data imbalance and semantic similarity between certain trademark categories continue to limit its effectiveness.

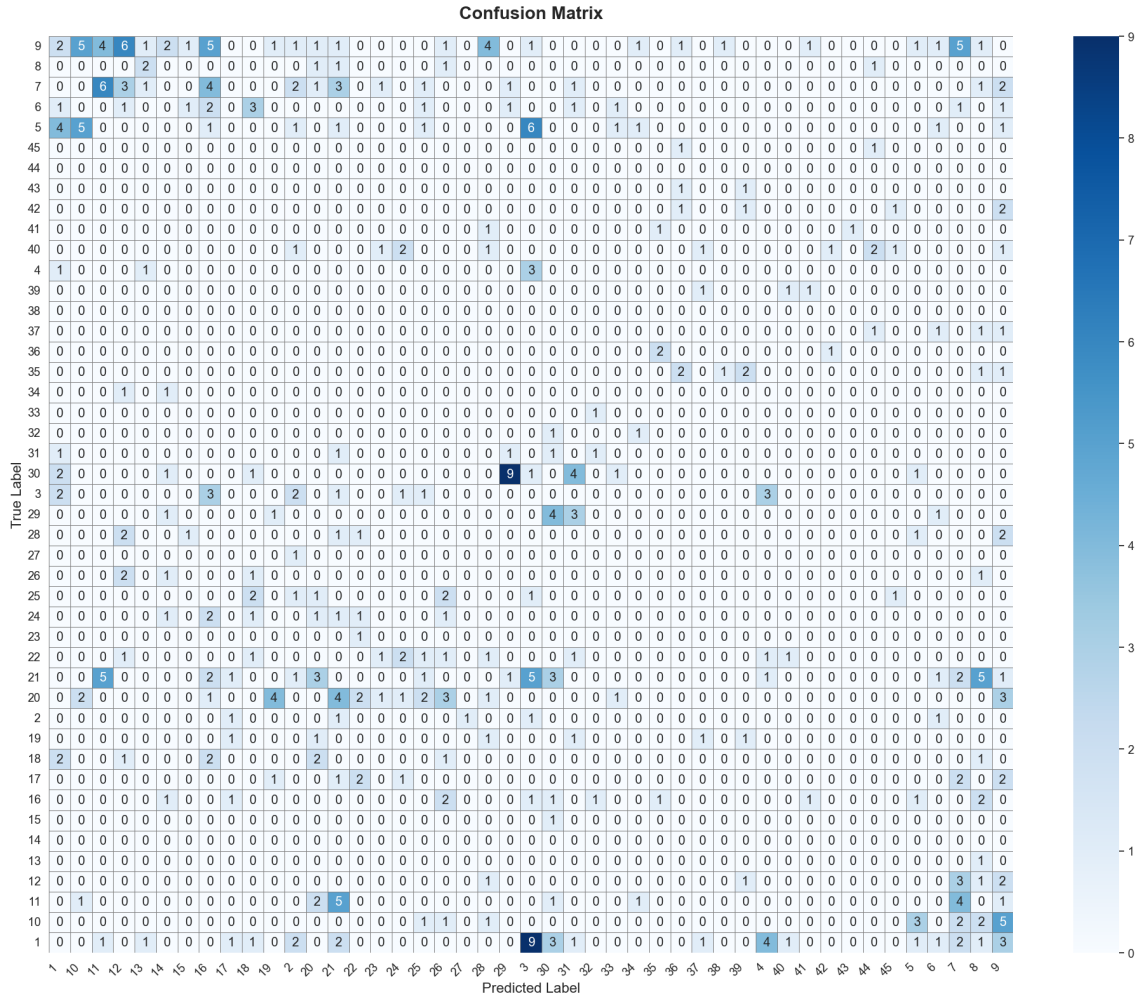


Figure 4.4: Off-diagonal confusion matrix after finetuning with class-weighted CE Loss.

These findings motivate the subsequent exploration of advanced training strategies and data augmentation techniques aimed at improving minority-class representation and reducing semantic overlap.

4.4 Two-stage Finetuning

4.4.1 Experiments

To address the severe class imbalance identified during the exploratory data analysis, a two-stage fine-tuning strategy was adapted and evaluated. The core idea of this approach is to stabilize the optimization process before introducing imbalance-aware learning signals.

In the first stage, the model is fine-tuned using class-weighted CE loss. This stage focuses on enabling the linear classification head to learn reliable decision boundaries while gradually adapting the higher-level representations of the encoder. During this stage, only the linear classification head and the last four Transformer layers of the encoder are unfrozen, while the remaining layers are kept frozen. The model is trained for 10 epochs to ensure stable convergence while accounting for class imbalance through normalized class weights.

In the second stage, the model is further fine-tuned using Focal Loss to explicitly address class imbalance by down-weighting easy majority-class samples and emphasizing harder and minority-class examples. The same unfreezing strategy is maintained, with the linear head and the last four encoder layers remaining trainable. This stage is also conducted for 10 epochs, allowing the model to refine its decision boundaries under imbalance-aware supervision.

To ensure a fair comparison, both single-stage baselines are trained for the same total number of epochs as the two-stage strategy. Specifically, the Single-stage class-weighted CE Loss and Single-stage Focal Loss models are fine-tuned for 20 epochs using the same unfreezing configuration, where only the linear classification head and the last four Transformer layers are trainable.

During training, model performance is monitored on the validation set at the end of each epoch. The checkpoint corresponding to the highest validation macro-averaged F1 score is selected and used for final evaluation on the test set for all compared fine-tuning strategies.

Table 4.4 reports the performance comparison between single-stage and two-stage fine-tuning strategies. Evaluation is performed using accuracy, macro-averaged F1 score, and weighted F1 score.

The experimental results demonstrate that the two-stage fine-tuning strategy

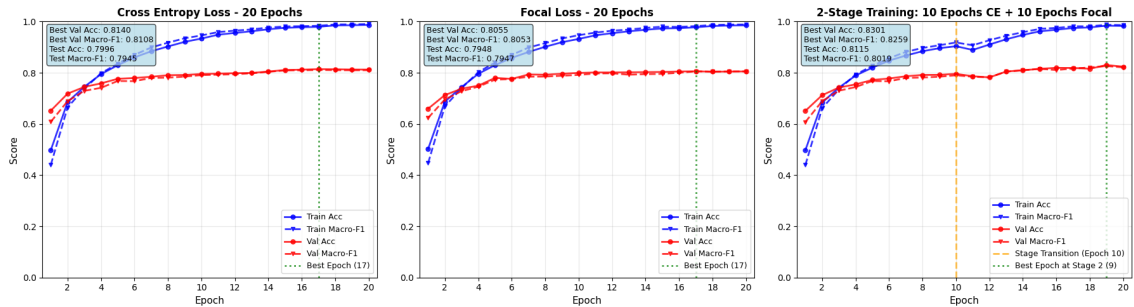
Table 4.4: Performance comparison of fine-tuning strategies

Training Strategy	Accuracy	Macro-F1	Weighted-F1
Single-stage class-weighted CE Loss	80.0%	79.5%	80.0%
Single-stage Focal Loss	79.5%	79.5%	79.5%
Two-stage (class-weighted CE $\rightarrow$ Focal)	81.1%	80.2%	81.1%

consistently outperforms single-stage training, even when all models are trained for an equal number of epochs. In particular, noticeable gains are observed in the macro-averaged F1 score, which increases from 79.5% with Single-stage class-weighted CE Loss to 80.2% with the two-stage approach. This improvement indicates enhanced recognition of minority classes and suggests that introducing Focal Loss after an initial stabilization phase with class-weighted CE Loss is more effective than applying imbalance-aware objectives throughout the entire training process.

4.4.2 Learning Curve Analysis

To understand the training dynamics and convergence behavior of the two-stage fine-tuning strategy, the learning curves for all three training configurations were examined. Figure 4.5 presents the training and validation performance trajectories over 20 epochs, comparing single-stage class-weighted Cross Entropy Loss, single-stage Focal Loss, and the two-stage approach.

**Figure 4.5:** Learning curves comparing single-stage and two-stage fine-tuning strategies.

The learning curves show that all three configurations learn quickly in the first 4-6 epochs. The two single-stage methods (class-weighted Cross Entropy and Focal Loss) have similar results, with both reaching their best validation performance at epoch 17 (0.8140 and 0.8155 accuracy, respectively).

The two-stage approach switches from class-weighted Cross Entropy to Focal Loss at epoch 10. After this switch, validation performance keeps improving and reaches its best at epoch 19 with 0.8301 accuracy and 0.8259 macro-F1. The two-stage method performs better than single-stage methods, even though all methods

train for the same number of epochs. The two-stage method also has better generalization, with a smaller difference between training and validation scores, showing that using loss functions one after another works better than using them separately.

4.4.3 Representation and Error Analysis

To demonstrate the performance improvement, the structural and classification characteristics of the two-stage model were analyzed using the test set.

Figure 4.6 presents the UMAP visualization of the sentence embeddings generated by the two-stage model. Compared to the single-stage class-weighted CE baseline, the cluster boundaries for several classes appear more distinct. However, semantic overlap persists in regions containing hard-to-distinguish categories, particularly those with limited training data.

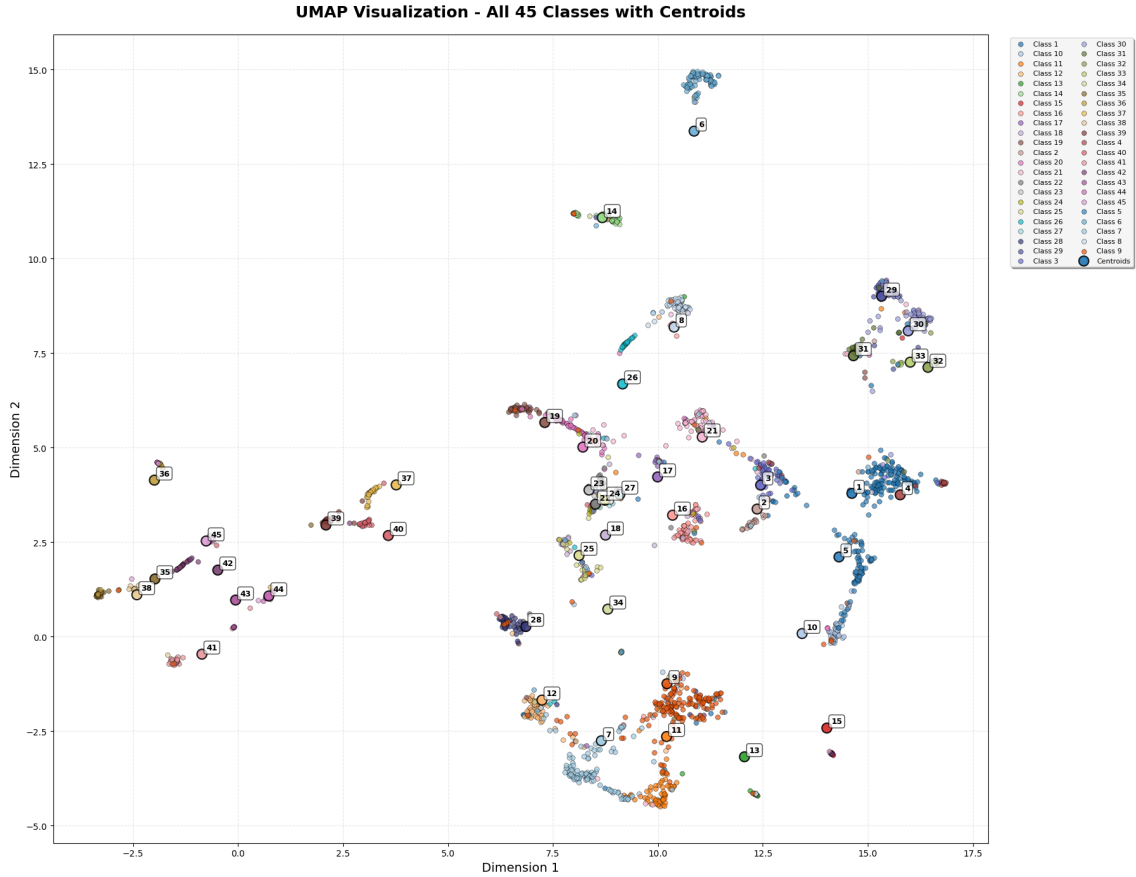


Figure 4.6: UMAP visualization of test set embeddings with two-stage fine-tuning.

Detailed performance metrics reveal that while the global metrics have improved, certain minority classes continue to pose significant challenges. Compared to the previous single-stage class-weighted CE configuration where sixteen classes exhibited at least one metric below 0.7, the two-stage strategy has reduced this number to ten classes: Classes 3, 4, 17, 18, 21, 22, 23, 24, 25, and 26. The analysis of these underperforming classes highlights two distinct error patterns. First, Class

17 (Precision=0.79, Recall=0.60) exhibits high precision but low recall, indicating that the model remains conservative in predicting this label, frequently misclassifying its samples into semantically dominant neighbor classes. In contrast, Class 22 (Precision=0.55, Recall=0.50) suffers from low values in both metrics, reflecting fundamental difficulty in distinguishing this category from semantically similar neighbors. Second, Class 23 exhibits a unique behavior with low precision (0.43) but relatively high recall (0.75). With a support size of only 4 samples in the test set, this volatility suggests that the model may have over-fitted to the few available examples during the Focal Loss stage, leading to a high false-positive rate. Similarly, Class 21 (Support=86) maintains a moderate F1 score of 0.69, indicating that simple re-weighting strategies are insufficient to fully resolve overlaps in medium-sized classes. Additionally, Class 3 shows a precision of only 0.62 despite higher recall (0.81), and Class 4 struggles with both precision (0.70) and recall (0.67), further demonstrating the persistent challenges of class imbalance.

Figure 4.7 illustrates the off-diagonal confusion matrix for this configuration. While the overall density of off-diagonal misclassifications has decreased compared to the baseline, systematic confusion remains between specific pairs of classes.

The persistence of low performance in extremely small classes (e.g., Class 23) despite the use of advanced loss functions and training strategies underscores a fundamental limitation: the lack of sufficient diverse training data. Loss re-weighting can only adjust the importance of existing samples but cannot generate new semantic information. This limitation motivates the final phase of the experimental pipeline, where LLM is employed to augment the training data, thereby explicitly injecting synthetic diversity into the minority classes.

4.5 Data Augmentation with LLM

To further enhance classification performance, particularly for underrepresented classes, a data augmentation strategy based on LLM was employed. Unlike conventional augmentation methods that rely on surface-level perturbations, the proposed approach focuses on the semantic refinement of trademark descriptions. The objective is to improve input quality and semantic clarity while strictly preserving the original class labels.

4.5.1 LLM-based Description Refinement

An instruction-tuned LLM was used to generate an improved description for each original trademark text. Rather than performing unrestricted paraphrasing, the generation process was guided by a carefully designed prompt that enforces semantic focus, stylistic consistency, and alignment with the NICE Classification

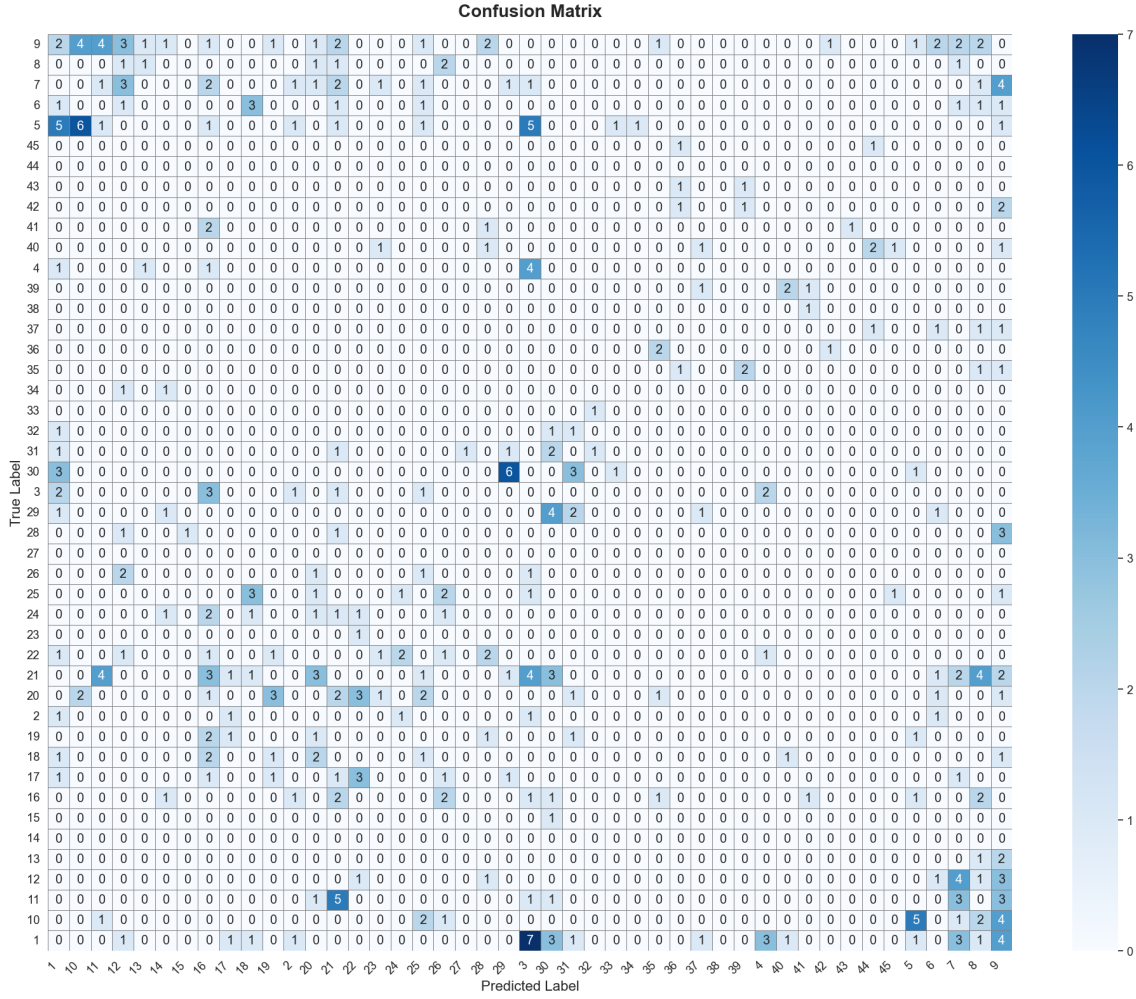


Figure 4.7: Off-diagonal confusion matrix with two-stage fine-tuning.

framework. This refinement process aims to reduce linguistic noise, normalize heterogeneous writing styles across languages, and explicitly highlight the core properties and functional intent of each product or service.

The prompt template used for data augmentation is shown below:

```
You are an expert in NICE Classification for trademarks.
Your task is to provide a concise definition of the prod-
uct/service based on the description below.
```

```
Requirements:
```

- Describe its nature (what it is/its properties) and purpose (primary function/utility)
- The output should be in {language} language only (strictly enforced)
- Maximum 50 words (strictly enforced)
- Use scientific and professional tone
- Do not include any redundant prefix or suffix in the output

```
Description: {description}
```

```
Output:
```

The prompt explicitly constrains output length, language, and stylistic properties, ensuring that the generated descriptions are concise, information-dense, and consistent across samples. Importantly, the semantic intent of the original description is preserved, while ambiguity and stylistic variation are significantly reduced. This results in a more standardized and discriminative textual representation that is better suited for downstream classification.

The LLM used for this augmentation process is *google/gemma-3n-E4B-it*, an instruction-tuned open-source model with approximately 8 billion parameters. Its strong instruction-following capability and multilingual understanding make it well-suited for domain-specific description refinement in a bilingual trademark classification setting.

4.5.2 Performance Impact of Data Augmentation

The LLM-refined descriptions were combined with the original training data and used to fine-tune the previously selected best-performing training configuration, namely the best BERT-based encoder with a linear classification head, [CLS] token representations, and the optimized loss function strategy identified in earlier experiments which is two-stage training with class-weighted CE and Focal Loss. Evaluation was conducted on the same held-out test set to ensure a fair and con-

trolled comparison.

Table 4.5 compares the classification performance before and after applying LLM-based data augmentation.

Table 4.5: Performance comparison before and after LLM-based data augmentation

Training Setting	Accuracy	Macro-F1	Weighted-F1
Before augmentation	81.1%	80.2%	81.1%
After LLM-based augmentation	83.6%	82.5%	83.6%

The results demonstrate a consistent improvement across all evaluation metrics after incorporating LLM-generated descriptions. The most pronounced gains are observed in the macro-averaged F1 score, indicating enhanced performance on minority classes. This suggests that the refined descriptions provide clearer semantic cues, enabling the model to better distinguish underrepresented categories that previously suffered from sparse and noisy textual signals.

4.5.3 Representation and Error Analysis

To verify the hypothesis that semantic refinement leads to better class separability, the geometric structure of the learned embeddings was analyzed. Figure 4.8 displays the UMAP projection of the test set embeddings after applying the LLM-based augmentation strategy. A qualitative comparison with previous visualizations reveals that the clusters are notably more compact, and the boundaries between semantically adjacent classes have become sharper. This indicates that the LLM-generated descriptions have effectively acted as a denoising mechanism, reducing intra-class variance and pushing samples closer to their respective class centroids.

A critical improvement is observed in the reduction of underperforming classes. In the previous two-stage fine-tuning phase, ten classes exhibited at least one metric below 0.7. Following data augmentation, this number has dropped to only two: Class 22 and Class 23. This reduction serves as strong evidence that semantic augmentation is particularly effective for minority classes where training data is scarce. Specifically, Class 23, which previously suffered from severe instability due to an extremely low support of 4 samples (Precision=0.43, F1=0.55), has shown substantial recovery. The precision for Class 23 increased to 0.60 while maintaining a high recall of 0.75, resulting in an F1-score of 0.67. This improvement suggests that the enriched descriptions helped the model reduce false positives by clarifying the specific definitions of this rare category.

Similarly, Class 22 demonstrated consistent gains across all metrics, with pre-

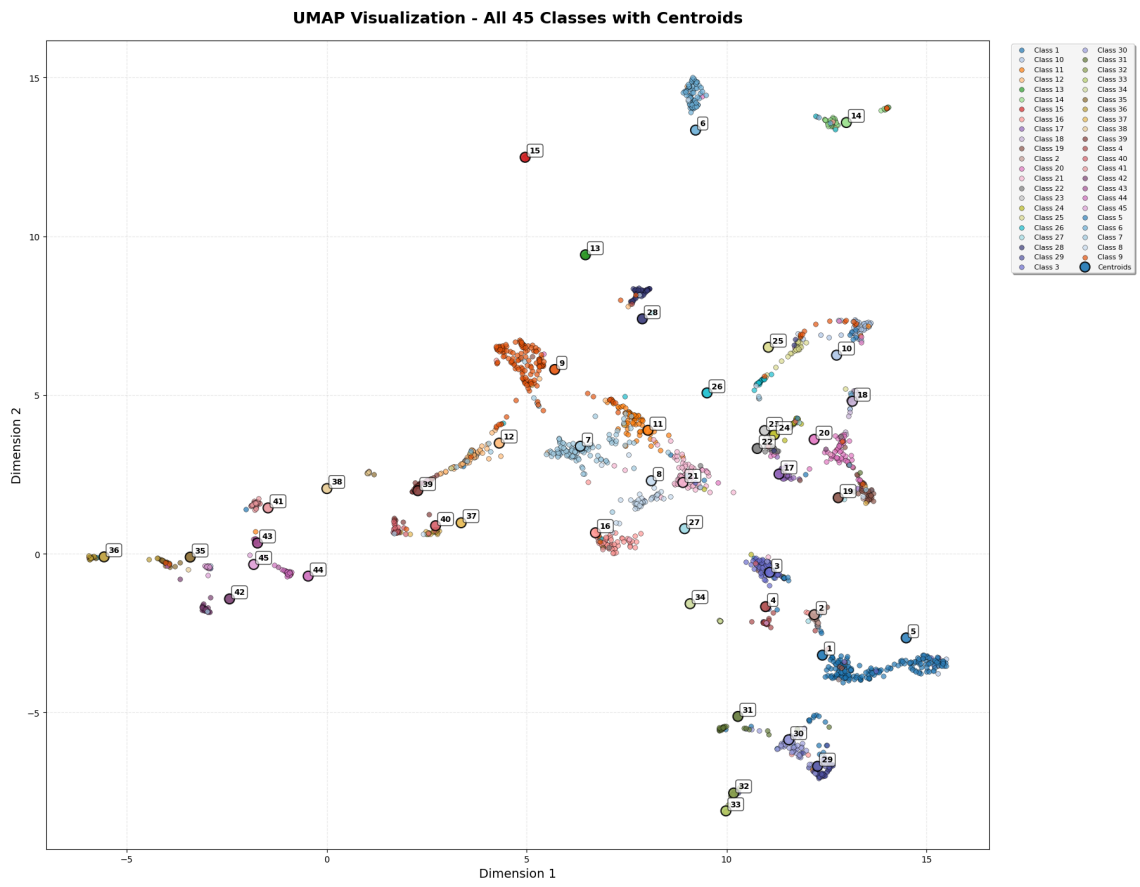


Figure 4.8: UMAP visualization of test set embeddings after LLM-based data augmentation.

cision increasing to 0.65 and recall to 0.59, yielding an F1-score of 0.62. Previously, this class struggled with low recall (0.50), implying that the model frequently missed relevant samples. The improvement in recall indicates that the augmented data provided more diverse yet semantically consistent examples, allowing the model to generalize better and retrieve more true positives. Although these two classes remain the most challenging, their absolute performance metrics are now significantly closer to the satisfactory threshold of 0.7.

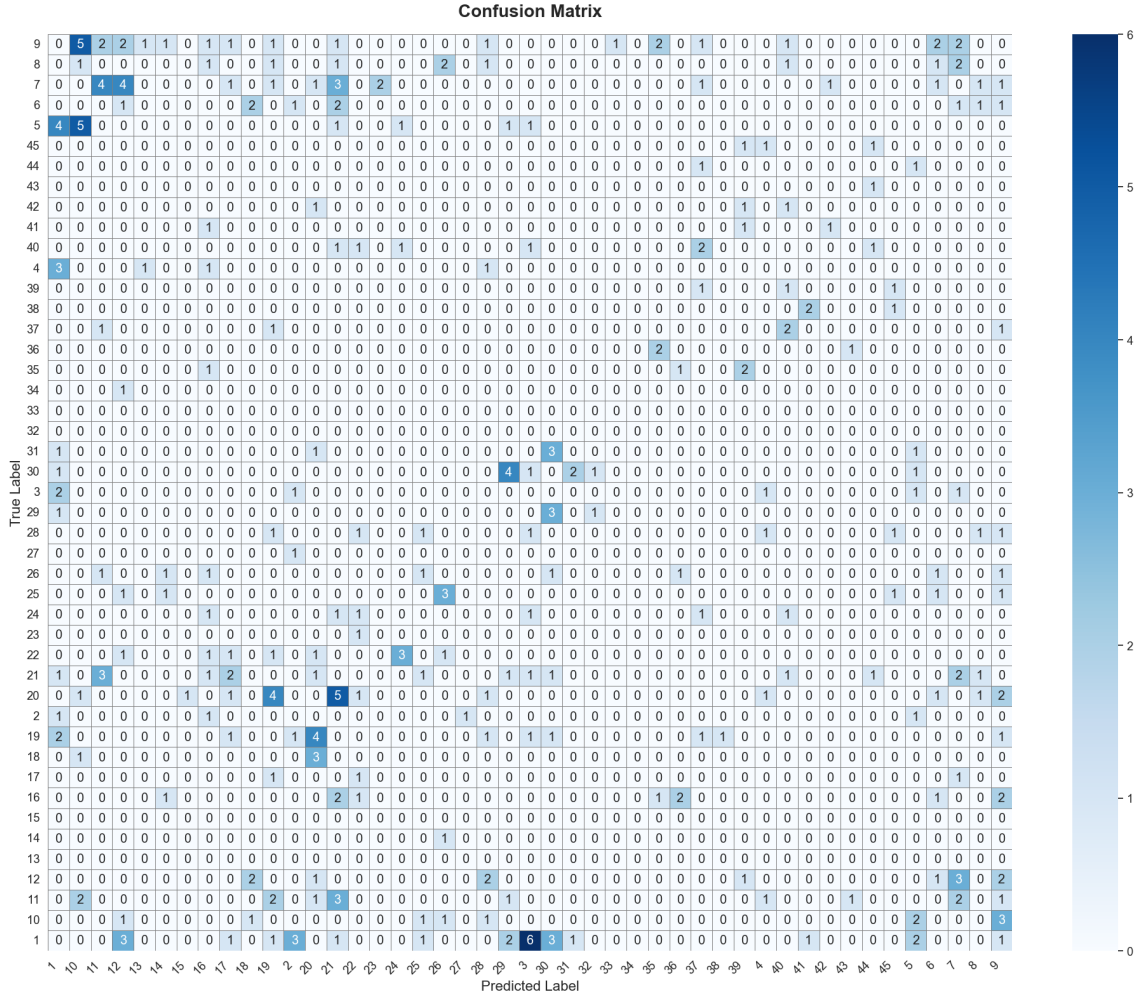


Figure 4.9: Off-diagonal confusion matrix after LLM-based data augmentation.

The reduction in systematic errors is further corroborated by the off-diagonal confusion matrix shown in Figure 4.9. The visualization displays a visibly sparser distribution of misclassifications compared to earlier experiments. The fading of previously dense off-diagonal regions confirms that the semantic overlap between confusing classes has been successfully mitigated. Overall, the analysis confirms that integrating LLM-generated knowledge into the training pipeline provides a robust solution for the trademark classification task under severe data imbalance.

CHAPTER 5. SOLUTION AND CONTRIBUTION

This chapter presents the main contributions of this research. The proposed methods solve the problem of multi-class text classification when data is very imbalanced, focusing on bilingual trademark classification.

5.1 Two-Stage Fine-Tuning Strategy

5.1.1 Problem Statement

In the trademark dataset, some classes have many samples while others have very few. The ratio between the largest and smallest classes is over 20:1. Some small classes have fewer than 50 training samples. This causes problems during training.

Standard CE Loss function treats all samples equally, so the model focuses too much on large classes and ignores small ones. Class-weighted CE Loss helps but still struggles with very small classes. Focal Loss [3] can help with imbalance, but using it from the start makes training unstable. When we tried Focal Loss alone, it performed worse than class-weighted CE Loss (79.5% vs. 80.0% accuracy). This led us to adapt the two-stage approach.

5.1.2 Proposed Solution

The solution uses two training stages. In the first stage, we use class-weighted CE Loss for 10 epochs. This helps the model learn stable decision boundaries and adapt to trademark descriptions. The model builds a good foundation.

In the second stage, we switch to Focal Loss [3] for another 10 epochs. By this time, the model already has good representations. Focal Loss then focuses more on hard examples from small classes and less on easy examples from large classes. This improves performance on small classes without breaking the overall training.

The key idea is that imbalance-aware training works better when the model starts from a good initial state. Using class-weighted CE Loss first, then Focal Loss, combines the benefits of both methods.

5.1.3 Results and Impact

As shown in Table 4.4, the two-stage method achieved 81.1% accuracy and 80.2% macro-F1, improving by 1.1 and 0.7 points over single-stage class-weighted CE Loss. The number of weak classes (with at least one metric below 0.7) dropped from 16 to 10, showing better performance on small classes.

5.2 LLM-Based Data Augmentation

5.2.1 Problem Statement

The two-stage method helped, but loss re-weighting only changes how we use existing data. It cannot add new information. Classes with very few samples, like Class 23 with only 4 test examples, still performed poorly.

Traditional augmentation methods like synonym replacement or back-translation change text at the surface level. They often lose important domain-specific meaning needed for trademark classification. They may also add noise that hurts performance.

Also, trademark descriptions in the dataset vary greatly in style and detail. Some are too long, others too short. The bilingual nature adds more variation between English and Vietnamese descriptions.

5.2.2 Proposed Solution

We use a LLM to refine trademark descriptions. Instead of just paraphrasing existing text, this method creates better versions that are more standardized and informative.

A prompt template guides the LLM to create refined descriptions that: describe the nature and purpose of each product or service, keep the same language as the original, stay under 50 words, use a professional tone, and avoid unnecessary words. This makes descriptions concise and useful for classification.

The refinement process has three benefits. First, it standardizes different writing styles, reducing variation within each class. Second, it highlights the key properties and purpose of each product or service. Third, it doubles the training data by adding refined versions, giving more examples to small classes.

We use *google/gemma-3n-E4B-it*, an open-source model with about 8 billion parameters. It follows instructions well and understands both English and Vietnamese.

5.2.3 Results and Impact

As shown in Table 4.5, the augmented model achieved 83.6% accuracy and 82.5% macro-F1, improving by 2.5 and 2.3 points over the non-augmented version. These gains are larger than from the two-stage method alone.

The biggest improvement is in reducing weak classes. Before augmentation, ten classes had at least one metric below 0.7. After augmentation, only two classes remained (Classes 22 and 23). Class 23 improved from precision 0.43 and F1 0.55

to precision 0.60 and F1 0.67. Class 22 improved from F1 0.52 to 0.62.

Analysis of the learned embeddings showed that class clusters became more compact after augmentation, with clearer boundaries between similar classes. This confirms that the LLM-generated descriptions reduced noise and variation within classes.

5.3 Combined Framework and Overall Performance

5.3.1 Integration of Contributions

The two methods work well together. The two-stage strategy solves optimization problems with imbalanced data, while LLM augmentation solves data scarcity by adding more examples.

When combined, they form a complete framework. First, we augment the original descriptions using LLM. Then, we train the model using the two-stage strategy: class-weighted CE Loss first, then Focal Loss.

5.3.2 Performance Summary

Table 5.1 shows the performance improvements from each step. The final system achieved 83.6% accuracy and 82.5% macro-F1, improving by 11.7 and 13.2 points over the baseline. The number of weak classes dropped from 16 to 2, showing that the methods effectively solved the class imbalance problem.

Table 5.1: Progressive performance improvements across experimental stages

Stage	Accuracy	Macro-F1	Weak Classes	Δ Macro-F1
Baseline (MPNet + SVM)	71.9%	69.3%	–	–
End-to-end MPNet (CE)	79.7%	79.1%	16	+9.8%
Two-stage MPNet (CE $\rightarrow$ Focal)	81.1%	80.2%	10	+1.1%
+ LLM Augmentation	83.6%	82.5%	2	+2.3%

These results show that each method addressed specific problems found in earlier stages, leading to cumulative improvements that were much better than using any single method alone.

CHAPTER 6. CONCLUSION AND FUTURE WORK

6.1 Conclusion

This thesis addressed the challenge of multi-class text classification under severe data imbalance, with a specific application to bilingual trademark classification according to the NICE Classification system. Through systematic experimentation and iterative refinement, a comprehensive framework was developed that significantly improves classification performance on both majority and minority classes.

6.1.1 Summary of Achievements

The research began with an exploratory data analysis that revealed the fundamental challenges of the trademark classification task. The dataset comprising 20,278 samples across 45 classes exhibited a highly skewed distribution, with imbalance ratios exceeding 20:1 between the largest and smallest categories. The bilingual composition of English and Vietnamese descriptions introduced additional complexity in semantic representation.

Baseline experiments established that pretrained multilingual BERT-based encoders substantially outperform traditional TF-IDF representations, with the BERT-based encoder achieving 71.9% accuracy compared to 62.0% for TF-IDF. The [CLS] token pooling strategy consistently yielded superior performance over mean pooling, leading to its adoption throughout subsequent experiments.

The transition from frozen encoder representations with SVM to end-to-end fine-tuning with a linear classification head resulted in a significant performance boost, increasing accuracy from 71.9% to 79.7%. This 7.8 percentage point improvement demonstrated the importance of allowing the pretrained encoder to adapt its representations to the domain-specific characteristics of trademark descriptions.

The proposed two-stage fine-tuning strategy, which combines CE Loss in the first stage with Focal Loss in the second stage, further improved performance to 81.1% accuracy and 80.2% macro-F1. This approach successfully addressed the optimization instability that occurs when applying imbalance-aware loss functions from the beginning of training. The number of classes with at least one metric below 0.7 decreased from 16 to 10, indicating meaningful improvements in minority-class performance.

Finally, the integration of LLM-based semantic augmentation yielded the most substantial single improvement, increasing accuracy to 83.6% and macro-F1 to 82.5%. By leveraging a carefully designed prompt to generate refined trademark

descriptions, the approach effectively doubled the training data while introducing semantic diversity that particularly benefited underrepresented classes. The number of underperforming classes decreased to only 2, with several previously challenging categories showing substantial recovery.

6.1.2 Comparison with Existing Approaches

The final system achieved a macro-F1 score of 82.5%, representing a total improvement of 13.2 percentage points over the baseline. This result demonstrates the effectiveness of the proposed framework for handling severe class imbalance in domain-specific text classification.

The two-stage fine-tuning strategy offers a principled alternative to single-stage training with imbalance-aware loss functions. Unlike approaches that apply Focal Loss from the beginning, the proposed method ensures stable optimization by first establishing reliable decision boundaries before introducing re-weighting. This insight has broader applicability beyond trademark classification to any multi-class task with significant imbalance.

The LLM-based semantic augmentation approach differs fundamentally from traditional text augmentation techniques. Rather than introducing surface-level perturbations that may corrupt domain-specific semantics, the proposed method generates standardized and information-dense descriptions that explicitly highlight classification-relevant features. This semantic refinement paradigm represents a novel application of generative language models to the data augmentation problem.

6.1.3 Limitations

Despite the significant improvements achieved, several limitations remain. First, two classes (Classes 22 and 23) continue to exhibit F1-scores below 0.7, indicating that the proposed approaches cannot fully resolve the challenges posed by extremely small sample sizes and high semantic similarity with neighboring categories.

Second, the LLM-based augmentation approach requires access to a capable language model, which may introduce computational overhead and dependency on external services. The quality of generated descriptions depends on the LLM's understanding of domain-specific terminology and the NICE Classification framework.

Third, the evaluation was conducted on a single dataset specific to the Vietnamese trademark classification context. While the proposed methods are designed

to be generalizable, their effectiveness on other languages, domains, or classification systems requires further validation.

6.2 Future Work

Several practical directions for future development emerge from this research.

First, Classes 22 and 23 still face classification challenges. These classes could be improved by using data augmentation methods to create more diverse training examples. Large language models could also be used to generate new synthetic training samples instead of only refining existing descriptions. This would help address the lack of data in minority classes.

Second, attention visualization methods could be added to help explain how the model makes decisions. This would make the classification results clearer and more useful for trademark examiners.

Finally, the model could be improved in real-world use by collecting and using feedback during operation.

In conclusion, this thesis has demonstrated that the combination of careful loss function scheduling and semantic data augmentation can substantially improve multi-class text classification under severe data imbalance. The proposed two-stage fine-tuning strategy and LLM-based semantic refinement approach represent generalizable contributions that extend beyond the specific application of trademark classification. The iterative experimental methodology adopted throughout this research provides a template for systematic model development in challenging real-world classification scenarios.

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APPENDIX

APPENDIX A. List of 45 NICE Classification Classes

This appendix presents the complete list of 45 classes in the NICE Classification system for trademarks, including both English and Vietnamese descriptions from IPVietnam.

1. Class 1

English: Chemicals for use in industry, science and photography, as well as in agriculture, horticulture and forestry; unprocessed artificial resins, unprocessed plastics; fire extinguishing and fire prevention compositions; tempering and soldering preparations; substances for tanning animal skins and hides; adhesives for use in industry; putties and other paste fillers; compost, manures, fertilizers; biological preparations for use in industry and science

Vietnamese: Hóa chất dùng trong công nghiệp, khoa học, nhiếp ảnh, cũng như trong nông nghiệp, nghề làm vườn và lâm nghiệp; Nhựa nhân tạo dạng thô, chất dẻo dạng thô; Hợp phần chữa cháy và phòng cháy; Chế phẩm dùng để tôi, ram và hàn kim loại; Chất để thuộc da sống và da động vật; Chất dính dùng trong công nghiệp; Mát tít và các loại bột nhào bít kín khác; Phân ú, phân chuồng, phân bón; Chế phẩm sinh học dùng trong công nghiệp và khoa học.

2. Class 2

English: Paints, varnishes, lacquers; preservatives against rust and against deterioration of wood; colorants, dyes; inks for printing, marking and engraving; raw natural resins; metals in foil and powder form for use in painting, decorating, printing and art

Vietnamese: Sơn, véc-ni, sơn mài; Chất chống rỉ và chất bảo quản gỗ; Chất nhuộm màu, màu nhuộm; Mực để in, đánh dấu và chạm khắc; Nhựa tự nhiên dạng thô; Kim loại dạng lá và dạng bột để sử dụng trong hội họa, trang trí, nghề in và nghệ thuật.

3. Class 3

English: Non-medicated cosmetics and toiletry preparations; nonmedicated dentifrices; perfumery, essential oils; bleaching preparations and other substances for laundry use; cleaning, polishing, scouring and abrasive preparations

Vietnamese: Mỹ phẩm và chế phẩm vệ sinh không chứa thuốc; Chế phẩm đánh răng không chứa thuốc; Nước hoa, tinh dầu; Chế phẩm tẩy trắng và các chất khác dùng để giặt; Chế phẩm để tẩy rửa, đánh bóng, tẩy dầu mỡ và mái mòn.

4. Class 4

English: Industrial oils and greases, wax; lubricants; dust absorbing, wetting and binding compositions; fuels and illuminants; candles and wicks for lighting

Vietnamese: Dầu và mỡ công nghiệp, sáp; Chất bôi trơn; Chất để thấm hút, làm ướt và làm dính bụi; Nhiên liệu và vật liệu cháy sáng; Nến và bấc dùng để thắp sáng.

5. Class 5

English: Pharmaceuticals, medical and veterinary preparations; sanitary preparations for medical purposes; dietetic food and substances adapted for medical or veterinary use, food for babies; dietary supplements for human beings and animals; plasters, materials for dressings; material for stopping teeth, dental wax; disinfectants; preparations for destroying vermin; fungicides, herbicides

Vietnamese: Các chế phẩm dược, y tế và thú y; Chế phẩm vệ sinh dùng cho mục đích y tế; Thực phẩm và chất dinh dưỡng phù hợp cho việc sử dụng trong y tế hoặc thú y, thực phẩm cho em bé; Chất bổ sung ăn kiêng cho người và động vật; Cao dân, vật liệu dùng để băng bó; Vật liệu để hàn răng, sáp nha khoa; Chất tẩy uế; Chất diệt động vật có hại; Chất diệt nấm, chất diệt cỏ.

6. Class 6

English: Common metals and their alloys, ores; metal materials for building and construction; transportable buildings of metal; non- electric cables and wires of common metal; small items of metal hardware; metal containers for storage or transport; safes

Vietnamese: Kim loại thường và quặng, hợp kim của chúng; Vật liệu xây dựng bằng kim loại; Cấu kiện bằng kim loại vận chuyển được; Cáp và dây kim loại thường không dùng để dẫn điện; Hàng ngũ kim và các vật dụng nhỏ làm bằng sắt; Đồ chứa đựng bằng kim loại để lưu giữ hoặc vận chuyển; Két sắt.

7. Class 7

English: Machines, machine tools, power-operated tools; motors and engines, except for land vehicles; machine coupling and transmission components, except for land vehicles; agricultural implements, other than hand-operated hand tools; incubators for eggs; automatic vending machines

Vietnamese: Máy, máy công cụ, dụng cụ vận hành bằng điện; Động cơ và đầu máy, trừ loại dùng cho phương tiện giao thông trên bộ; Các bộ phận ghép nối và truyền động, trừ loại dùng cho phương tiện giao thông trên bộ; Nông cụ, trừ loại

dụng cụ cầm tay thao tác thủ công; Máy ấp trứng; Máy bán hàng tự động.

8. Class 8

English: Hand tools and implements, hand-operated; cutlery; side arms, except firearms; razors

Vietnamese: Công cụ và dụng cụ cầm tay thao tác thủ công; Dao, kéo, thìa và đĩa; Vũ khí đeo cạnh sườn, trừ súng cầm tay; Dao cao.

9. Class 9

English: Scientific, research, navigation, surveying, photographic, cinematographic, audiovisual, optical, weighing, measuring, signalling, detecting, testing, inspecting, life-saving and teaching apparatus and instruments; apparatus and instruments for conducting, switching, transforming, accumulating, regulating or controlling the distribution or use of electricity; apparatus and instruments for recording, transmitting, reproducing or processing sound, images or data; recorded and downloadable media, computer software, blank digital or analogue recording and storage media; mechanisms for coinoperated apparatus; cash registers, calculating devices; computers and computer peripheral devices; diving suits, divers' masks, ear plugs for divers, nose clips for divers and swimmers, gloves for divers, breathing apparatus for underwater swimming; fire-extinguishing apparatus

Vietnamese: Thiết bị và dụng cụ khoa học, nghiên cứu, hàng hải, trắc địa, nhiếp ảnh, điện ảnh, nghe nhìn, quang học, cân, đo, báo hiệu, dò tìm, kiểm tra, giám sát, cấp cứu và giảng dạy; Thiết bị và dụng cụ dùng để truyền dẫn, chuyển mạch, biến đổi, tích, điều chỉnh hoặc điều khiển sự phân phối hoặc sử dụng năng lượng điện; Thiết bị hoặc dụng cụ dùng để ghi, truyền, tái tạo hoặc xử lý âm thanh, hình ảnh hoặc dữ liệu; Các phương tiện chứa dữ liệu và có thể tái xuống được, phần mềm máy tính, các phương tiện ghi và lưu trữ dữ liệu kỹ thuật số hoặc tương tự chưa có dữ liệu; Cơ cấu cho thiết bị vận hành bằng đồng xu; Máy tính tiền, máy tính; máy vi tính và thiết bị ngoại vi của máy vi tính; Bộ quần áo lặn, mặt nạ lặn, tai nghe dùng cho thợ lặn, kẹp mũi dùng cho thợ lặn và người bơi lội, găng tay dùng cho thợ lặn, thiết bị thờ dùng để bơi; Thiết bị dập lửa.

10. Class 10

English: Surgical, medical, dental and veterinary apparatus and instruments; artificial limbs, eyes and teeth; orthopaedic articles; suture materials; therapeutic and assistive devices adapted for persons with disabilities; massage apparatus; apparatus, devices and articles for nursing infants; sexual activity apparatus, devices and articles

Vietnamese: Thiết bị và dụng cụ phẫu thuật, y tế, nha khoa và thú y; Chăn, tay, mắt và răng giả; Dụng cụ chỉnh hình; Vật liệu khâu vết thương; Thiết bị trợ giúp và trị liệu dùng cho người tàn tật; Thiết bị xoa bóp; Thiết bị, dụng cụ và vật dụng để chăm sóc trẻ sơ sinh; Thiết bị, dụng cụ và vật dụng kích thích hoạt động tình dục.

11. Class 11

English: Apparatus and installations for lighting, heating, cooling, steam generating, cooking, drying, ventilating, water supply and sanitary purposes

Vietnamese: Thiết bị và hệ thống để chiếu sáng, sưởi nóng, làm mát, sinh hơi nước, nấu nướng, sấy khô, thông gió, cấp nước và dùng cho mục đích vệ sinh.

12. Class 12

English: Vehicles; apparatus for locomotion by land, air or water

Vietnamese: Xe cộ; Phương tiện giao thông trên bộ, trên không hoặc dưới nước.

13. Class 13

English: Firearms; ammunition and projectiles; explosives; fireworks

Vietnamese: Súng cầm tay; Đạn dược và đầu đạn; Chất nổ; Pháo hoa.

14. Class 14

English: Precious metals and their alloys; jewellery, precious and semiprecious stones; horological and chronometric instruments

Vietnamese: Kim loại quý và hợp kim của chúng; Đồ trang sức, đồ kim hoàn, đá quý và bàn quý; Đồng hồ và dụng cụ đo thời gian.

15. Class 15

English: Musical instruments; music stands and stands for musical instruments; conductors' batons

Vietnamese: Dụng cụ âm nhạc; Giá đỡ bản nhạc và giá giữ nhạc cụ; Que đũa đánh nhịp của nhạc trưởng.

16. Class 16

English: Paper and cardboard; printed matter; bookbinding material; photographs; stationery and office requisites, except furniture; adhesives for stationery or household purposes; drawing materials and materials for artists; paintbrushes; instructional and teaching materials; plastic sheets, films and bags for wrapping and packaging; printers' type, printing blocks

Vietnamese: Giấy, bìa cứng; Ấn phẩm; Vật liệu để đóng sách; Ảnh chụp; Văn

phòng phẩm và đồ dùng văn phòng, trừ đồ đặc; Keo dán dùng cho văn phòng hoặc dùng cho mục đích gia dụng; Vật liệu dùng để vẽ và vật liệu dùng cho nghệ sĩ; Bút lông; Đồ dùng để hướng dẫn và giảng dạy; Tấm, màng mỏng, túi bằng chất dẻo dùng để bọc và bao gói; Chữ in, bàn in đúc (clisé).

17. Class 17

English: Unprocessed and semi-processed rubber, gutta-percha, gum, asbestos, mica and substitutes for all these materials; plastics and resins in extruded form for use in manufacture; packing, stopping and insulating materials; flexible pipes, tubes and hoses, not of metal

Vietnamese: Cao su, nhựa pec-ca, gôm, amiang, mi-ca bán thành phẩm và chưa xử lý và các vật liệu thay thế cho những vật liệu này; Chất dẻo và nhựa ở dạng đùn ép, sử dụng trong sản xuất; Vật liệu để bao gói, bịt kín và cách ly; Ống mềm phi kim loại.

18. Class 18

English: Leather and imitations of leather; animal skins and hides; luggage and carrying bags; umbrellas and parasols; walking sticks; whips, harness and saddlery; collars, leashes and clothing for animals

Vietnamese: Da và giả da; Da động vật và da sống; Túi hành lý và túi xách; Ổ và dù; Gậy chống đi bộ; Roi ngựa và yên cương; Vòng cổ, dây buộc và trang phục cho động vật.

19. Class 19

English: Materials, not of metal, for building and construction; rigid pipes, not of metal, for building; asphalt, pitch, tar and bitumen; transportable buildings, not of metal; monuments, not of metal

Vietnamese: Vật liệu, không bằng kim loại, dùng trong xây dựng; Ống cứng, phi kim loại, dùng cho xây dựng; Asphan, hắc ín, nhựa đường và bitum; Công trình xây dựng vận chuyển được, phi kim loại; Đài kỷ niệm, phi kim loại.

20. Class 20

English: Furniture, mirrors, picture frames; containers, not of metal, for storage or transport; unworked or semi-worked bone, horn, whalebone or mother-of-pearl; shells; meerschaum; yellow amber

Vietnamese: Đồ đạc, gương, khung tranh; Thùng chứa, không bằng kim loại, dùng để lưu trữ hoặc vận chuyển; Xương, sừng, phiến sừng ở hàm cá voi hoặc xà cừ dạng thô hoặc bán thành phẩm; Vỏ sò, vỏ ốc; Đá bọt; Hồ phách vàng.

21. Class 21

English: Household or kitchen utensils and containers; cookware and tableware, except forks, knives and spoons; combs and sponges; brushes, except paintbrushes; brush-making materials; articles for cleaning purposes; unworked or semi-worked glass, except building glass; glassware, porcelain and earthenware

Vietnamese: Dụng cụ và đồ chứa dùng cho gia đình hoặc bếp núc; Dụng cụ nấu nướng và bộ đồ ăn, trừ đĩa, dao và thìa; Lược và bọt biển; Bàn chải, trừ chổi quét sơn/bút lông để vẽ; Vật liệu dùng để làm bàn chải; Vật dụng dùng để làm sạch; Thủy tinh thô hoặc bán thành phẩm, trừ kính xây dựng; Đồ thủy tinh, sành, sứ và đất nung.

22. Class 22

English: Ropes and string; nets; tents and tarpaulins; awnings of textile or synthetic materials; sails; sacks for the transport and storage of materials in bulk; padding, cushioning and stuffing materials, except of paper, cardboard, rubber or plastics; raw fibrous textile materials and substitutes therefor

Vietnamese: Dây và dây thừng; Lưới; Lều (trại) và vải dầu/vải nhựa/vải bạt; Mái che bằng chất liệu dệt hoặc tổng hợp; Buồm; Bao bì đựng để vận chuyển và lưu giữ nguyên liệu dạng rời; Vật liệu để đệm (lót) và nhồi, trừ giấy, bìa cứng, cao su hoặc chất dẻo; Vật liệu sợi dệt dạng thô và vật liệu thay thế chúng.

23. Class 23

English: Yarns and threads for textile use

Vietnamese: Các loại sợi dùng để dệt

24. Class 24

English: Textiles and substitutes for textiles; household linen; curtains of textile or plastic

Vietnamese: Vải và hàng dệt; Vải lanh dùng trong nhà; Rèm che bằng vải dệt hoặc bằng chất dẻo.

25. Class 25

English: Clothing, footwear, headwear

Vietnamese: Quần áo, đồ đi chân, đồ đội đầu

26. Class 26

English: Lace, braid and embroidery, and haberdashery ribbons and bows; buttons, hooks and eyes, pins and needles; artificial flowers; hair decorations; false

hair

Vietnamese: Đăng ten, dải và đồ để thêu trang trí, ruy băng và nơ dùng để may vá; Khuy, ghim móc và khuyết, kẹp và kim khâu; Hoa nhân tạo; Đồ trang trí cho tóc; Tóc giả.

27. Class 27

English: Carpets, rugs, mats and matting, linoleum and other materials for covering existing floors; wall hangings, not of textile

Vietnamese: Thảm, chiếu, thảm chùi chân, vải sơn và các vật liệu trải sàn khác, giấy dán tường, không dệt.

28. Class 28

English: Games, toys and playthings; video game apparatus; gymnastic and sporting articles; decorations for Christmas trees

Vietnamese: Trò chơi, đồ chơi; Thiết bị chơi trò chơi video; Dụng cụ thể dục thể thao; Đồ trang hoàng cây Noel.

29. Class 29

English: Meat, fish, poultry and game; meat extracts; preserved, frozen, dried and cooked fruits and vegetables; jellies, jams, compotes; eggs; milk, cheese, butter, yogurt and other milk products; oils and fats for food

Vietnamese: Thịt, cá, gia cầm và thú săn; Chất chiết ra từ thịt; Rau, quả được bảo quản, phơi khô hoặc nấu chín; Nước quả nấu đông, mứt ứt, mứt quả ứt; Trứng; Sữa, pho mát, bơ, sữa chua và các sản phẩm sữa khác; Dầu thực vật và mỡ dùng cho thực phẩm

30. Class 30

English: Coffee, tea, cocoa and substitutes therefor; rice, pasta and noodles; tapioca and sago; flour and preparations made from cereals; bread, pastries and confectionery; chocolate; ice cream, sorbets and other edible ices; sugar, honey, treacle; yeast, baking-powder; salt, seasonings, spices, preserved herbs; vinegar, sauces and other condiments; ice (frozen water)

Vietnamese: Cà phê, trà (chè), ca cao và các sản phẩm thay thế chúng; Gạo, mì ống và mì sợi; Bột sắn bột và bột cọ sagu; Bột và chế phẩm làm từ ngũ cốc; Bánh mì, bánh ngọt và bánh kẹo; Sô cô la; Kem lạnh, kem trái cây và các loại đá lạnh khác ăn được; Đường, mật ong, nước mật đường; Men, bột nở; Muối, gia vị, thảo mộc đã bảo quản; Dấm, nước xốt và các loại gia vị khác; Kem (nước đông lạnh).

31. Class 31

English: Raw and unprocessed agricultural, aquacultural, horticultural and forestry products; raw and unprocessed grains and seeds; fresh fruits and vegetables, fresh herbs; natural plants and flowers; bulbs, seedlings and seeds for planting; live animals; foodstuffs and beverages for animals; malt

Vietnamese: Các sản phẩm nông nghiệp, nuôi trồng thủy sản, làm vườn và lâm nghiệp dạng thô và chưa xử lý; Các loại ngũ cốc và hạt dạng thô và chưa xử lý; Rau, củ và quả tươi, thảo mộc tươi; Cây và hoa tự nhiên; Củ, cây con và hạt giống để trồng; Động vật sống; Thức ăn và đồ uống cho động vật; Mạch nha.

32. Class 32

English: Beers; non-alcoholic beverages; mineral and aerated waters; fruit beverages and fruit juices; syrups and other preparations for making non-alcoholic beverages

Vietnamese: Bia; Đồ uống không có cồn; Nước khoáng và nước uống có ga; Đồ uống hoa quả và nước ép trái cây; Xi-rô và các chế phẩm khác để làm đồ uống không có cồn.

33. Class 33

English: Alcoholic beverages, except beers; alcoholic preparations for making beverages

Vietnamese: Đồ uống có cồn, trừ bia; Các chế phẩm có cồn dùng để làm đồ uống.

34. Class 34

English: Tobacco and tobacco substitutes; cigarettes and cigars; electronic cigarettes and oral vaporizers for smokers; smokers' articles; matches

Vietnamese: Thuốc lá và các sản phẩm thay thế thuốc lá; Thuốc lá điếu và xì gà; Thuốc lá điện tử và bình sinh hơi cho người hút thuốc; Vật dụng cho người hút thuốc; Diêm.

35. Class 35

English: Advertising; business management, organization and administration; office functions

Vietnamese: Quảng cáo; Quản lý, tổ chức và quản trị kinh doanh; Hoạt động văn phòng.

36. Class 36

English: Financial, monetary and banking services; insurance services; real estate services

Vietnamese: Dịch vụ tài chính, tiền tệ và ngân hàng; Dịch vụ bảo hiểm; Bất động sản.

37. Class 37

English: Construction services; installation and repair services; mining extraction, oil and gas drilling

Vietnamese: Dịch vụ xây dựng; Dịch vụ lắp đặt và sửa chữa; Dịch vụ khai thác mỏ, khoan dầu và khí.

38. Class 38

English: Telecommunications services

Vietnamese: Dịch vụ viễn thông.

39. Class 39

English: Transport; packaging and storage of goods; travel arrangement

Vietnamese: Vận tải; Đóng gói và lưu giữ hàng hoá; Du lịch.

40. Class 40

English: Treatment of materials; recycling of waste and trash; air purification and treatment of water; printing services; food and drink preservation

Vietnamese: Xử lý vật liệu; Tái chế chất thải và rác thải; Lọc không khí và xử lý nước; Dịch vụ in ấn; Bảo quản thực phẩm và đồ uống.

41. Class 41

English: Education; providing of training; entertainment; sporting and cultural activities

Vietnamese: Giáo dục; Đào tạo; Giải trí; Tổ chức các hoạt động thể thao và văn hoá.

42. Class 42

English: Scientific and technological services and research and design relating thereto; industrial analysis, industrial research and industrial design services; quality control and authentication services; design and development of computer hardware and software

Vietnamese: Dịch vụ khoa học và công nghệ và nghiên cứu và thiết kế liên quan đến chúng; Dịch vụ nghiên cứu công nghiệp, dịch vụ phân tích công nghiệp

và dịch vụ thiết kế công nghiệp; Dịch vụ kiểm soát và xác thực chất lượng; Thiết kế và phát triển phần cứng và phần mềm máy tính.

43. Class 43

English: Services for providing food and drink; temporary accommodation

Vietnamese: Dịch vụ cung cấp thực phẩm và đồ uống; Chỗ ở tạm thời.

44. Class 44

English: Medical services; veterinary services; hygienic and beauty care for human beings or animals; agriculture, aquaculture, horticulture and forestry services

Vietnamese: Dịch vụ y tế; Dịch vụ thú y; Chăm sóc vệ sinh và sắc đẹp cho người hoặc động vật; Dịch vụ nông nghiệp, nuôi trồng thủy sản, dịch vụ làm vườn và dịch vụ lâm nghiệp.

45. Class 45

English: Legal services; security services for the physical protection of tangible property and individuals; dating services, online social networking services; funerary services; babysitting

Vietnamese: Dịch vụ pháp lý; Dịch vụ an ninh thực thể nhằm bảo vệ người và tài sản hữu hình; Dịch vụ hẹn hò, dịch vụ mạng xã hội trực tuyến; Dịch vụ tang lễ; Dịch vụ trông giữ trẻ tại nhà.